A project report on

An Accurate Object Tracking Framework for Drone-Based Marine Surveillance

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology in Computer Science Engineering

by

JEFFERSON DAVID KINGSTON (21BCE1882)

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Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

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I hereby declare that the thesis entitled "An Accurate Object Tracking Framework for Drone-Based Marine Surveillance" submitted by Jefferson David Kingston (21BCE1882), for the award of the degree of Bachelor of Technology in Computer Science and Engineering, Vellore Institute of Technology, Chennai is a record of bonafide work carried out by me under the supervision of Dr. Ancy Micheal.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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Date: 22/11/2024

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Signature of the Candidate

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CERTIFICATE

This is to certify that the report entitled "An Accurate Object Tracking Framework for Drone-Based Marine Surveillance" is prepared and submitted by **Jefferson David Kingston (21BCE1882)** to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science Engineering** is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

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ABSTRACT

This project presents an advanced framework for accurate human detection and tracking in marine surveillance applications using Unmanned Aerial Vehicles (UAVs). UAVs are increasingly deployed for crucial operations such as human rescue, anti-poaching, and environmental monitoring in vast and challenging marine environments. Given the complex dynamics of marine environments, effective tracking of human activities is critical for safety, conservation, and regulatory enforcement. The proposed framework integrates YOLOv8 for object detection with DeepSORT for object tracking and association, achieving state-of-the-art accuracy in marine scenarios.

The methodology uses YOLOv8 for robust object detection, capable of handling small spatial objects and challenging orientations typical of aerial marine videos. YOLOv8's advanced architecture incorporates modules like the Cross Stage Partial Network (CSPNet) and attention mechanisms such as CBAM, enabling it to focus on key elements in cluttered or dynamic scenes. Following detection, DeepSORT leverages a Kalman filter for predictive tracking and the Hungarian algorithm for object association, assigning unique IDs to individuals across video frames for uninterrupted tracking. This approach significantly enhances tracking reliability, even in crowded or overlapping areas.

Our framework was evaluated on the MOBDrone dataset, which includes 66 videos with over 126,000 frames captured at altitudes between 10-60 meters above sea level. The dataset comprises various classes, including persons, surfboards, boats, wood, and lifebuoys. Precision, Recall, and F-Score were used as performance metrics, with YOLOv8 + DeepSORT achieving Recall of 88.7%, Precision of 92.9%, and F1-Score of 90.75%, outperforming other models such as SSD, DSOD, and previous YOLO iterations. These results underscore the model's effectiveness for real-time aerial surveillance, meeting the high precision and speed demands essential for maritime rescue and anti-poaching efforts.

This framework's innovative combination of YOLOv8 and DeepSORT not only improves tracking accuracy but also enhances emergency response capabilities in marine settings. By reliably identifying and tracking individuals, it supports conservationists, researchers, and rescue teams in rapid decision-making, making it a valuable tool for aerial-based marine surveillance.

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Date: 22/11/2024 Jefferson David Kingston

CONTENTS

CHAPTER 1

INTRODUCTION..1

1.1 BACKGROUND..1

1.2 PROBLEM STATEMENT 2

1.3 RESEARCH OBJECTIVES 4

1.4 SIGNIFICANCE OF THE STUDY 5

1.5 SCOPE AND LIMITATIONS 6

CHAPTER 2

LITERATURE REVIEW

2.1 UAV TECHNOLOGY IN MARINE APPLICATION 8

2.2 OBJECT DETECTION FRAMEWORKS 9

2.3 OBJECT TRACKING SYSTEMS 10

2.4 MARINE SURVEILLANCE SYSTEMS 12

2.5 RESEARCH GAP ANALYSIS..13

CHAPTER 3

RESEARCH METHODOLOGY

3.1 SYSTEM ARCHITECTURE..15

3.2 YOLOV8 IMPLEMENTATION……………………………………..16

3.3 DEEPSORT INTEGRATION………………………………………..19

CHAPTER 4

RESULTS AND ANALYSIS

4.1 PERFORMANCE METRICS 22

4.2 COMPARATIVE ANALYSIS…………………………………………24

4.3 DATASET DESCRIPTION……………………………………………26

4.4 EXPERIMENTAL SETUP……………………………………….……….30

CHAPTER 5

APPLICATIONS AND CASE STUDIES

5.1 MARINE LIFE PROTECTION..34

5.2 SEARCH AND RESCUE OPERATIONS……………………………...36

5.3 MARITIME LAW ENFORCEMENT………………………………….37

5.4 ENVIRONMENTAL MONITORING………………………………….39

CHAPTER 6

DISCUSSION

6.1 TECHNICAL IMPLICATIONS...42

6.2 PRACTICAL IMPLICATIONS 44

6.3 LIMITATIONS AND CHALLENGES 46

6.4 FUTURE IMPROVEMENTS...49

CHAPTER 7

CONCLUSION AND FUTURE WORK 52

APPENDICES 54

1. DATASET AND PRE PROCESSING
2. MODEL CONFIGURATION AND EXPERIMENTAL SETUP
3. RESULTS AND ERROR ANALYSIS
4. OUTPUT

REFERENCES..59

LIST OF FIGURES

1. CIB ARCHITECTURE 17
2. PSA ARCHITECTURE 18
3. PERFORMANCE METRICS OF DIFFERENT METHODS 24
4. SAMPLE DATASET IMAGES 27
5. OBJECT TRACKING ASSOCIATION IN MOBDRONE DATASET 58

LIST OF TABLES

1. COMPARATIVE ANALYSIS OF VARIOUS MODELS 53

LIST OF ACRONYMS

YOLO	You only Look Once
CIB	Compact Inverted Block
PSA	Partial Self Attention
UAV	Unnamed Aerial Vehicle
IoU	Intersection-over-Union
DeepSORT	Deep Simple Online and Realtime Tracking

Chapter 1

Introduction

1.1 BACKGROUND

1.1.1 Evolution of UAV Technology

The evolution of Unmanned Aerial Vehicles (UAVs) in marine surveillance represents a significant technological leap in maritime monitoring capabilities. Initially developed for military applications in the 1960s, UAVs have undergone remarkable transformations over the past six decades. The first generation of UAVs was primarily focused on basic aerial photography and reconnaissance missions, with limited payload capacity and flight duration. The 1990s marked a crucial turning point with the introduction of GPS navigation systems and improved battery technology, enabling more precise flight control and extended operational ranges.

The early 2000s witnessed a paradigm shift with the integration of advanced sensors and imaging systems. High-resolution cameras, thermal imaging capabilities, and multispectral sensors expanded the UAVs' application scope in marine environments. The development of more efficient propulsion systems and lightweight materials significantly enhanced flight endurance and payload capacity. Modern UAVs now incorporate sophisticated features such as obstacle avoidance systems, autonomous flight capabilities, and real-time data transmission, making them increasingly suitable for complex marine surveillance operations.

1.1.2 Current State of Marine Surveillance

Contemporary marine surveillance has evolved into a sophisticated ecosystem where UAVs play a pivotal role alongside traditional monitoring methods. Current UAV-based surveillance systems employ a multi-layered approach, combining real-time video feeds, automated detection algorithms, and intelligent tracking systems. These systems operate at various altitudes, from low-level coastal monitoring to high-altitude open ocean surveillance, providing unprecedented coverage and flexibility.

The integration of artificial intelligence and machine learning algorithms has revolutionized data processing capabilities. Modern systems can automatically detect and classify various marine objects, from small vessels to marine mammals, with increasing accuracy. Real-time object tracking algorithms, enhanced by deep learning techniques, enable continuous monitoring of targets of interest. The current generation of UAVs also features advanced communication systems that facilitate seamless data transmission to ground control stations, enabling immediate response to detected anomalies or emergencies.

1.1.3 Challenges in Maritime Monitoring

Despite significant technological advancements, maritime monitoring faces several persistent challenges that impact surveillance effectiveness. Environmental factors pose primary concerns, with varying weather conditions significantly affecting UAV operations. High winds, precipitation, and salt spray can compromise sensor performance and flight stability. The dynamic nature of marine environments, characterized by constantly moving waves and changing light conditions, presents unique challenges for object detection and tracking algorithms.

Technical limitations also remain significant hurdles. Battery life constraints continue to restrict continuous monitoring capabilities, particularly in remote ocean areas where charging facilities are unavailable. Communication range limitations pose challenges for real-time data transmission, especially in areas beyond coastal networks. The need for robust and weather-resistant hardware adds to system complexity and operational costs. Additionally, regulatory frameworks governing UAV operations in maritime spaces often impose operational restrictions that must be carefully navigated.

The processing of vast amounts of surveillance data presents another significant challenge. High-resolution imagery and video feeds require substantial computational resources for real-time analysis. The need to maintain high accuracy in object detection while operating in real-time creates a delicate balance between processing speed and detection precision. Furthermore, the integration of multiple data streams from various sensors and the fusion of this information for meaningful analysis adds another layer of complexity to maritime monitoring systems.

Operational challenges extend to the coordination of UAV surveillance with existing maritime traffic and activities. The need to maintain safe distances from vessels, aircraft, and marine infrastructure while conducting surveillance operations requires sophisticated flight planning and control systems. The development of reliable collision avoidance mechanisms and integration with maritime traffic management systems remains an ongoing challenge in the field.

1.2 PROBLEM STATEMENT

1.2.1 NEED FOR ACCURATE OBJECT TRACKING IN MARINE ENVIRONMENT

The marine environment presents unique and complex challenges for object tracking systems, particularly in the context of UAV-based surveillance. Unlike land-based surveillance, marine environments are characterized by constantly moving backgrounds due to wave action, varying sea states, and dynamic weather conditions. This environmental complexity significantly impacts the accuracy and reliability of object detection and tracking systems. The need for precise object tracking becomes particularly critical in applications such as search and rescue operations, where even slight inaccuracies could have severe consequences.

The maritime domain requires tracking systems capable of handling multiple object types simultaneously, from small objects like floating debris and people in water to larger vessels and marine mammals. These objects often exhibit similar visual characteristics, making accurate classification and continuous tracking extremely challenging. Furthermore, the reflective nature of water surfaces creates specular highlights and glare that can interfere with optical sensors, potentially leading to false detections or tracking failures. The system must maintain high accuracy despite these adverse conditions, as the stakes in marine surveillance operations are often life-critical.

1.2.2 LIMITATIONS OF CURRENT SURVEILLANCE SYSTEMS

Current maritime surveillance systems suffer from several significant limitations that hamper their effectiveness in real-world applications. Traditional radar-based systems, while effective for larger vessels, often struggle to detect and track smaller objects, especially in rough sea conditions. Satellite-based surveillance systems, though offering wide coverage, are constrained by their temporal resolution and weather dependencies. These systems typically provide periodic snapshots rather than continuous monitoring, creating potential gaps in surveillance coverage during critical events.

The integration of existing systems poses another major limitation. Many current solutions operate in isolation, lacking the capability to seamlessly share data and coordinate responses. This fragmentation leads to inefficiencies in monitoring and response times. Additionally, most existing systems are designed for specific types of targets or conditions, lacking the flexibility to adapt to varying surveillance requirements. The high cost of deployment and maintenance of traditional surveillance infrastructure further limits the feasibility of comprehensive coverage, particularly in remote maritime regions.

1.2.3 TECHNICAL CHALLENGES IN UAV-BASED TRACKING

UAV-based tracking systems face numerous technical challenges that must be addressed to achieve reliable maritime surveillance. One primary challenge is the limited computational resources available on UAV platforms. The need for real-time processing of high-resolution video feeds while maintaining tracking accuracy places significant demands on onboard processors. This creates a constant trade-off between processing power, battery life, and system performance. The integration of sophisticated tracking algorithms must be optimized to operate within these hardware constraints without compromising accuracy.

Communication bandwidth limitations present another significant technical hurdle. The transmission of high-quality video feeds and sensor data in real-time requires substantial bandwidth, which may not always be available in maritime environments. The system must be capable of maintaining tracking functionality even when facing communication constraints or temporary loss of connection with ground control stations. Additionally, the development of robust tracking algorithms that can handle varying lighting conditions, weather changes, and different sea states while maintaining consistent performance remains a significant technical challenge.

The requirement for long-duration operations adds another layer of complexity to UAV-based tracking systems. The need to balance power consumption between various system components - propulsion, sensors, processing units, and communication systems - while maintaining optimal tracking performance presents a significant engineering challenge. Furthermore, the system must be designed to handle environmental factors such as wind gusts, electromagnetic interference, and salt spray, which can affect both the UAV platform and its sensor systems. These technical challenges must be addressed comprehensively to develop a reliable and effective maritime surveillance solution.

1.3 RESEARCH OBJECTIVE

1.3.1 PRIMARY OBJECTIVES

The primary objective of this research is to develop a robust object detection and tracking framework tailored for Unmanned Aerial Vehicles (UAVs) deployed in complex marine environments. UAVs have proven invaluable in various domains of surveillance and monitoring, but their application in marine settings brings a unique set of challenges. Detecting and tracking objects, particularly humans or vessels, from aerial views above water presents issues such as motion blur, fluctuating light conditions, and environmental noise like ocean waves. This research seeks to address these challenges by creating an accurate and efficient tracking system specifically optimized for marine surveillance. By focusing on enhanced human detection and tracking capabilities, the project aims to improve response time and precision in applications such as marine life rescue operations, poaching prevention, environmental monitoring, and disaster response scenarios. This framework's primary goal is to ensure a reliable solution that operates effectively in real-time, maintaining tracking accuracy even in high-variance marine conditions, ultimately serving as a tool for safer and more informed decision-making in ocean surveillance.

1.3.2 SPECIFIC GOALS

To achieve these primary objectives, the project sets forth several specific goals. Firstly, it aims to employ and fine-tune the latest YOLOv8 model for high-resolution object detection, taking advantage of its advanced feature extraction and optimized architecture designed for smaller, irregularly shaped objects, which are often encountered in marine aerial views. YOLOv8's improvements in handling low-resolution and partially obscured objects make it particularly suitable for marine applications. The second goal is to integrate DeepSORT as the tracking algorithm, leveraging its capability for robust and continuous tracking by using both appearance descriptors and motion models. DeepSORT's use of Kalman filtering enables predictive tracking, helping maintain object continuity even during partial occlusions, while the Hungarian algorithm ensures that each detected object is uniquely and consistently tracked across frames. Furthermore, this research seeks to optimize these algorithms to minimize computational overhead, thus achieving real-time tracking capabilities essential for UAV deployment in the field. The third goal is to validate this framework on the MOBDrone dataset, a dataset specifically designed to simulate the challenging aspects of maritime environments, using quantitative metrics such as precision, recall, and F-Score. These specific goals are aimed at creating a balanced and highly functional system that is both computationally efficient and adaptable to the nuanced demands of marine surveillance.

1.3.3 EXPECTED OUTCOMES

The expected outcomes of this research are multifaceted, encompassing both technical and practical advancements. Technically, the project is anticipated to produce a high-precision UAV-based tracking system capable of accurately detecting and following human figures or other critical objects across diverse and unpredictable marine environments. Specifically, the project targets achieving benchmark performance metrics with a recall rate of 88% or higher, precision at 92% or higher, and an F-Score around 90%. Such performance metrics would make this system a leading solution in the domain of UAV marine surveillance, meeting the industry standards for accuracy and reliability in high-stakes applications. Practically, this system is expected to streamline operations in marine environments by allowing operators to quickly and accurately locate and track subjects of interest, significantly improving the efficiency of rescue and enforcement operations. Additionally, the system's real-time tracking capabilities would make it ideal for deployment in situations requiring immediate response, such as accident monitoring, wildlife protection, or disaster management. Beyond its initial deployment, the research framework established here is expected to serve as a foundation for further enhancements, including the integration of additional environmental sensors, more advanced machine learning models, and extended functionalities like automated alerting systems. In the broader context, this research contributes to the growing field of UAV-based monitoring and AI-driven environmental surveillance, providing a scalable solution with the potential to significantly enhance operational outcomes in various marine and coastal applications.

1.4 SIGNIFICANCE OF THE STUDY

1.4.1 IMPACT ON MARINE CONSERVATION

This study has the potential to play a crucial role in marine conservation efforts by providing a reliable and efficient system for tracking and monitoring activities in marine environments. As human activities increasingly threaten marine ecosystems through illegal fishing, pollution, and other harmful practices, a robust surveillance system that operates effectively from an aerial perspective is essential. By accurately detecting and tracking vessels and human activity, this research offers a practical solution for identifying and deterring harmful practices in real-time. The UAV-based framework developed in this project will empower conservation authorities to monitor vast marine areas without the need for constant human presence, thus reducing operational costs and increasing coverage. Additionally, this system's ability to distinguish between different types of objects, such as boats, marine animals, or people, could enable targeted interventions, helping authorities respond promptly to potential threats to marine biodiversity. Consequently, this research supports sustainable marine resource management by providing conservationists with the tools necessary for proactive monitoring and preservation efforts.

1.4.2 APPLICATIONS IN RESCUE OPERATIONS

The study also holds significant implications for enhancing search and rescue operations in marine settings. With the challenges of unpredictable weather, changing light conditions, and vast, open water, traditional rescue methods are often limited in scope and response time. This UAV-based tracking system offers a real-time solution that enables rescuers to monitor and detect stranded individuals, vessels in distress, or other emergency scenarios with improved accuracy and efficiency. The system's adaptability to different weather and environmental conditions ensures a high degree of reliability, critical for time-sensitive rescue missions. By leveraging the YOLOv8 detection model, the framework can effectively identify and locate individuals or vessels even under challenging visual conditions, while the DeepSORT tracking algorithm provides consistent tracking even when subjects are temporarily obscured. This capability reduces search times and enhances situational awareness for first responders, ultimately leading to faster, more effective rescues and potentially saving lives. In addition to immediate rescue applications, this technology could assist in disaster response scenarios, such as identifying and tracking survivors during natural disasters like hurricanes or floods, making it a versatile tool for emergency response agencies.

1.4.3 CONTRIBUTION TO TECHNOLOGICAL ADVANCEMENTS

Beyond its direct applications in marine conservation and rescue, this research contributes to the broader field of technological innovation, particularly in aerial surveillance and AI-driven object tracking. The integration of advanced object detection models like YOLOv8 with optimized tracking algorithms such as DeepSORT represents a step forward in creating high-performance, adaptable UAV systems capable of operating autonomously in complex environments. This research paves the way for further advancements in aerial-based monitoring systems, offering insights into efficient processing, adaptive tracking, and real-time data handling. Additionally, the project's focus on overcoming environmental challenges associated with marine settings—such as motion blur, lighting changes, and partial occlusions—enhances the adaptability of UAV-based tracking systems, setting a benchmark for future developments in similar domains. The methodological advancements achieved in this project, from optimizing computational efficiency to enhancing detection precision, contribute valuable knowledge to the fields of computer vision and aerial robotics. The study thus serves as a foundation for future research, encouraging further exploration into sophisticated, AI-driven monitoring solutions for environmental and humanitarian applications worldwide.

1.5 SCOPE AND LIMITATIONS

1.5.1 GEOGRAPHICAL SCOPE

The project primarily targets marine environments where aerial surveillance can have a significant impact on monitoring, conservation, and rescue operations. The system is designed for applications in diverse marine geographies, including coastal zones, open seas, and areas with heavy maritime traffic, such as shipping routes or fishing hotspots. While the focus is on oceanic and near-shore environments, the adaptability of the system allows for use across various bodies of water, enabling monitoring in high-biodiversity regions and areas with critical conservation needs. However, the UAV-based solution faces limitations in extremely remote or expansive regions, where extended battery life, long-range data transmission, and navigation may become challenges. Additionally, harsh weather conditions and high waves common to certain maritime areas can potentially impede visibility and impact the UAV's tracking capabilities, restricting the system's effectiveness in such environments.

1.5.2 TECHNICAL BOUNDARIES

This project leverages advanced AI models like YOLOv8 for object detection and DeepSORT for tracking, both of which offer state-of-the-art accuracy and efficiency. However, the system's technical performance may vary depending on environmental factors like lighting, object occlusion, and turbulence. While the YOLOv8 model is optimized for high-speed and high-accuracy detection, the system may struggle in low-visibility conditions, such as heavy fog, dusk, or glare from the sun reflecting off the water. Additionally, the size and positioning of the UAV can influence detection range and resolution, potentially limiting its capability to track smaller or partially obscured objects. Processing constraints within UAV hardware further limit the complexity of onboard AI models, potentially impacting real-time analysis speed. Moreover, the AI system’s detection algorithms might require recalibration when encountering unfamiliar or rapidly changing marine scenes, limiting the universality of the solution across different marine contexts.

1.5.3 IMPLEMENTATION CONSTRAINTS

Several logistical and regulatory constraints impact the practical implementation of this UAV-based tracking system. UAV operation in maritime environments often requires compliance with stringent aviation and marine regulations, particularly in restricted or international waters. Navigating these regulatory frameworks can pose challenges for full-scale deployment, especially in areas with no-fly zones or where UAV usage is limited to specific operational guidelines. Additionally, current UAV battery technology limits operational flight time, which can be a significant constraint for extended monitoring missions. Battery life may require frequent recharges or replacements, potentially increasing costs and reducing monitoring continuity. Another limitation is the reliance on stable GPS signals and reliable data transmission to ground stations or cloud platforms, which may be impacted by the vast and remote nature of some marine locations. Finally, the costs associated with UAV maintenance, regular software updates for AI models, and potential data storage needs can restrict accessibility for some organizations, particularly those with limited resources or in developing regions, where budget constraints could limit full system integration. Despite these constraints, the project lays foundational groundwork to address these limitations over time through technological advancements, regulatory adaptations, and alternative energy solutions.

Chapter 2

Literature Review

2.1 UAV TECHNOLOGY IN MARINE APPLICATIONS

2.1.1 HISTORICAL DEVELOPMENT

The use of Unmanned Aerial Vehicles (UAVs) in marine applications has evolved significantly over recent decades. Initially developed for military and surveillance purposes, UAV technology has gradually expanded into commercial and environmental fields, with marine applications emerging as a critical area of interest. Early marine UAVs were primarily deployed for surface-level reconnaissance in maritime defense, assisting in monitoring illegal fishing, border control, and search and rescue missions. Over time, advancements in UAV hardware and software enhanced flight stability, battery life, and data-capturing capabilities, which expanded their applicability in non-military marine sectors. These advancements allowed researchers and conservationists to deploy UAVs for monitoring marine biodiversity, mapping coastal habitats, and conducting oceanographic surveys, setting the stage for broader and more specialized applications in the field of marine surveillance.

2.1.2 CURRENT TRENDS

Today, UAV technology in marine applications is heavily focused on integrating artificial intelligence (AI) and machine learning (ML) for real-time data processing and decision-making. Many UAVs are equipped with sophisticated sensors, including high-resolution cameras, thermal imaging, and LiDAR, enabling them to capture detailed data across various environmental conditions. Additionally, AI-driven object detection and tracking algorithms have become essential for analyzing marine environments, identifying endangered species, and detecting illegal activities such as poaching or unregulated fishing. Recent trends also include deploying autonomous UAV swarms that work in coordinated groups to cover larger areas efficiently, which is particularly useful for wide-scale environmental monitoring and disaster management. Furthermore, UAVs are increasingly being used in hybrid operations, complementing data from other sources, such as underwater drones and satellite imagery, to build comprehensive and multi-dimensional views of marine ecosystems.

2.1.3 FUTURE DIRECTIONS

The future of UAV technology in marine applications holds promising advancements aimed at overcoming current limitations. With ongoing research in battery technology, UAVs are expected to achieve longer flight times, expanding their operational range for continuous surveillance missions. Additionally, future UAVs may incorporate solar-powered or hybrid engines to enable sustainable and extended missions, especially in remote or open-sea environments. AI algorithms are anticipated to become more sophisticated, with improved capabilities in predictive analytics, enabling UAVs to anticipate and track movements based on historical data and environmental patterns. Emerging technologies like quantum computing could further enhance the processing power available for real-time data analysis, allowing for more accurate object detection and tracking under challenging conditions. Furthermore, advancements in communication technology, including 5G and beyond, are expected to support more stable and efficient data transmission, crucial for long-distance maritime operations. These developments are likely to broaden UAV applications in marine conservation, climate monitoring, and search and rescue missions, marking a significant technological shift toward more autonomous, intelligent, and resilient UAV systems in marine environments.

2.2 OBJECT DETECTION FRAMEWORKS

2.2.1 TRADITIONAL METHODS

Traditional object detection methods primarily relied on handcrafted features and classical machine learning techniques for identifying objects within images or videos. Early approaches such as the use of edge detection algorithms, histogram of oriented gradients (HOG), and scale-invariant feature transform (SIFT) were effective in specific controlled environments but had significant limitations in handling complex or dynamic scenes. These methods typically required extensive pre-processing, where the object of interest had to be manually segmented or isolated, followed by feature extraction. Classical machine learning algorithms such as Support Vector Machines (SVM) or k-Nearest Neighbors (k-NN) were then used to classify the objects based on the extracted features. However, these traditional methods struggled with large datasets, variation in object appearance, and real-time performance, leading to the development of more sophisticated approaches that could handle a variety of challenging conditions, such as varying lighting, occlusions, and diverse object shapes.

2.2.2 DEEP LEARNING APPROACHES

The advent of deep learning significantly transformed object detection by automating the feature extraction process through convolutional neural networks (CNNs). Deep learning models, particularly CNNs, are designed to automatically learn hierarchical features from data without requiring manual intervention. This approach has substantially outperformed traditional methods, especially in complex environments where objects may appear in varied sizes, orientations, or lighting conditions. Modern deep learning-based frameworks for object detection, such as Faster R-CNN, R-FCN, and SSD (Single Shot Multibox Detector), combine the power of CNNs with region proposal networks (RPNs) to identify and classify objects in images with impressive accuracy and speed. These methods significantly reduced the dependency on feature engineering and provided better results in tasks involving real-time video processing, large-scale data, and multi-class object recognition. Additionally, deep learning-based object detection frameworks became more efficient at generalizing across various environments and conditions, making them suitable for a wide range of applications, from surveillance and medical imaging to autonomous vehicles and marine monitoring.

2.2.3 YOLO FAMILY EVOLUTION

Among the many deep learning-based object detection frameworks, the You Only Look Once (YOLO) family has become one of the most popular and widely used models due to its balance between accuracy and real-time performance. The YOLO model takes a unique approach by framing object detection as a single regression problem, predicting both class labels and bounding box coordinates in one pass through the network. This results in faster processing times, making YOLO ideal for real-time applications.

YOLO's initial versions, such as YOLOv1, demonstrated a breakthrough in terms of speed, but faced challenges in detecting small objects and handling overlapping objects. However, each subsequent version of YOLO has seen notable improvements in both speed and accuracy. YOLOv2 (also known as Darknet-19) introduced better performance by utilizing batch normalization and multi-scale training, improving detection for smaller objects and more complex scenes. YOLOv3, with its more refined architecture, incorporated multi-scale prediction and better handling of smaller objects by using a deeper backbone network. The current evolution of YOLO, YOLOv7, continues to push the boundaries of performance by integrating new advancements such as more efficient architectures, improved bounding box prediction techniques, and better handling of occlusions and diverse object shapes. The YOLO family's continuous evolution has made it the framework of choice for real-time object detection in applications like autonomous driving, security surveillance, and marine monitoring, where fast and accurate detection is critical. With each iteration, YOLO models are becoming more adaptable and efficient, solidifying their role in modern computer vision tasks.

2.3 OBJECT TRACKING SYSTEMS

2.3.1 CLASSICAL TRACKING METHODS

Classical object tracking methods emerged as early solutions to maintain the identity of objects over time within a sequence of images or video frames. These methods often relied on low-level image features, such as edges, corners, or color histograms, to track objects. A well-known early technique is the Kalman filter, which uses statistical models to predict the future state of an object based on its current position and motion. It works by assuming the object follows a linear path and is useful for tracking objects with constant velocity in ideal conditions. Other classical methods, such as optical flow and mean-shift tracking, use variations in pixel intensities or patterns of movement to predict an object's motion across frames. However, classical tracking methods often struggle when objects undergo significant changes in appearance, experience occlusions, or interact with other objects in complex environments. Furthermore, these methods tend to be computationally expensive and require extensive fine-tuning to achieve reliable tracking under various conditions.

2.3.2 MODERN TRACKING ALGORITHMS

Modern tracking algorithms have evolved significantly to address the limitations of classical methods, particularly in dynamic or cluttered environments. One of the key advancements in this area has been the incorporation of deep learning techniques. Unlike classical methods, which rely on manually defined features, modern tracking algorithms leverage the power of convolutional neural networks (CNNs) to automatically extract robust and complex features for tracking. One notable example is the Correlation Filter-based Tracking algorithm, which aims to correlate the object with its template across multiple frames using a filter that is learned during tracking. Algorithms such as Siamese networks, which are trained to learn similarity metrics between two image patches, have also become highly effective for real-time tracking tasks. These modern methods can handle larger and more complex variations in object appearance, such as scaling, rotation, and lighting conditions. Another important development in modern tracking is the integration of multi-object tracking (MOT), which enables the simultaneous tracking of multiple objects in a scene, addressing scenarios where more than one object needs to be monitored concurrently. These modern methods have led to significant advancements in applications such as autonomous driving, video surveillance, and even robotics, where real-time tracking of multiple moving objects is crucial.

2.3.3 DEEPSORT AND ALTERNATIVES

Among the modern object tracking algorithms, DeepSORT (Deep Learning-based SORT) has gained significant attention due to its ability to track multiple objects accurately and in real-time. DeepSORT is an extension of the SORT (Simple Online and Realtime Tracking) algorithm, which itself is a lightweight and efficient tracker that uses Kalman filters and the Hungarian algorithm for data association. DeepSORT enhances the SORT framework by integrating a deep learning-based appearance descriptor to improve its robustness in handling object occlusions and re-identifications. The key strength of DeepSORT lies in its ability to track objects across frames even when they are temporarily occluded or re-enter the scene after being lost. This is achieved by combining motion information from Kalman filtering with appearance information obtained through a deep convolutional neural network, making it highly effective in dynamic and crowded environments.

There are several alternatives to DeepSORT, each tailored to specific tracking scenarios. For instance, Tracktor (Tracking by Regression) simplifies the tracking process by using regression networks to directly predict the position and bounding box of an object. FairMOT (Fair Multi-Object Tracking) is another promising alternative that combines both detection and tracking into a single framework, making it efficient for tracking multiple objects in real-time with consistent accuracy. Unlike DeepSORT, FairMOT addresses the issue of object identity switches by introducing an integrated approach that uses a shared backbone network to handle both tasks. Other alternatives include ByteTrack, which focuses on reducing the computational complexity of multi-object tracking without sacrificing accuracy, and CenterTrack, which simplifies the process of object tracking by associating keypoint locations in a centralized manner.

The ongoing development of these tracking algorithms reflects the growing need for highly efficient and accurate object tracking in real-world applications, including autonomous vehicles, video surveillance, and marine monitoring. As the performance of these algorithms continues to improve, their ability to handle increasingly complex environments and scenarios will be crucial for the success of many computer vision-based applications.

2.4 MARINE SURVEILLANCE SYSTEMS

2.4.1 EXISTING FRAMEWORKS

Marine surveillance systems have been largely based on traditional technologies such as radar, satellite imagery, and the Automatic Identification System (AIS) for detecting and tracking maritime vessels. Radar systems are highly effective for tracking large vessels over significant distances and in various weather conditions. Satellite imaging is used for broad environmental surveillance and can cover vast areas, but it lacks the real-time, high-resolution capability needed for dynamic surveillance. AIS, which provides data on the identity, location, and course of vessels equipped with transponders, is useful for tracking larger commercial vessels but is not capable of detecting unregistered vessels or smaller craft. To overcome these limitations, more recent developments have integrated Unmanned Aerial Vehicles (UAVs) and Unmanned Underwater Vehicles (UUVs) into marine surveillance. UAVs equipped with high-resolution cameras, infrared sensors, and advanced computing capabilities allow for real-time aerial monitoring of marine environments. UAVs are increasingly being used for monitoring marine life, tracking vessels, and supporting search and rescue operations. UUVs, on the other hand, are designed for underwater surveillance, detecting illegal fishing activities, or exploring underwater habitats. Together, UAVs and UUVs significantly enhance the capability of marine surveillance systems.

2.4.2 COMPARITIVE ANALYSIS

When compared to traditional surveillance methods, UAVs offer clear advantages in terms of flexibility, resolution, and real-time data acquisition. While radar systems can monitor large areas and provide valuable data about vessel movements, UAVs are far superior in their ability to provide high-resolution, real-time images and videos of both large and small objects, including vessels without AIS transponders. Satellites, though effective for large-scale environmental monitoring, cannot compete with UAVs in terms of timely updates or the ability to operate in localized, specific areas. UAVs also have a distinct edge in tracking smaller vessels or monitoring areas that are remote or have little to no coverage from conventional methods. Despite these advantages, UAVs are limited by flight time, battery life, and sensitivity to adverse weather conditions such as strong winds and rain, which can affect their operational capacity. In contrast, AIS-based systems cannot detect vessels that do not have an AIS transponder, highlighting a key area where UAVs fill an important gap. The integration of UAVs with advanced detection systems (like machine learning-based object recognition) and sensors enhances their tracking ability, overcoming some of the inherent challenges faced by traditional technologies.

2.4.3 PERFORMANCE METRICS

The performance of marine surveillance systems is often evaluated using several key metrics, including accuracy, precision, recall, real-time processing capability, coverage area, robustness, and cost-effectiveness. Accuracy is crucial for ensuring that vessels are correctly identified and tracked, while precision and recall measure the system's ability to minimize false positives and false negatives. UAV systems, in particular, are assessed on how well they track and identify objects in real-time, which is essential for time-sensitive operations such as search and rescue. The coverage area refers to the extent of marine territory a surveillance system can effectively monitor, and UAVs, depending on their endurance and range, can offer a flexible solution for both short-range and long-range surveillance. Real-time processing is necessary to provide actionable insights during emergencies or rapid events, while the robustness of the system evaluates its ability to function under varying weather conditions and environmental factors. Finally, cost-effectiveness is an important consideration when integrating UAVs into marine surveillance operations, as their operational costs, including maintenance and technology upgrades, need to be balanced with the benefits of enhanced monitoring. As UAV technology advances, the performance of marine surveillance systems continues to improve, with better integration of AI, machine learning, and sensor technologies allowing for more accurate, efficient, and reliable marine monitoring solutions.

2.5 RESEARCH GAP ANALYSIS

2.5.1 CURRENT LIMITATIONS

While significant progress has been made in marine surveillance systems, several key limitations still hinder their effectiveness. Traditional systems, such as radar and AIS, offer valuable tracking capabilities but are limited in their ability to detect smaller, unregistered vessels or objects. Radar systems, for example, struggle with small objects such as marine debris or smaller boats, and AIS relies heavily on the voluntary installation of transponders on vessels, leaving out many potential targets. Furthermore, current surveillance systems often suffer from limited real-time processing capabilities, making them less effective for rapid response situations such as search and rescue operations. UAVs have been increasingly adopted in marine surveillance to overcome some of these limitations, but challenges remain regarding flight time, weather dependence, and payload capacity. UAVs also require highly advanced image processing algorithms to achieve accurate object detection in complex marine environments, which can be a limiting factor. In addition, current tracking methods, particularly classical tracking algorithms, struggle with the dynamic and unpredictable nature of marine environments, which can lead to errors in object identification and tracking.

2.5.2 OPPORTUNITIES FOR IMPROVEMENT

There are several areas where marine surveillance systems can be significantly improved. One of the major opportunities lies in enhancing UAV-based systems, particularly in their ability to operate in adverse weather conditions and extend flight times. Battery technology and autonomous flight algorithms can be further developed to improve endurance, enabling UAVs to remain operational for longer durations, which is crucial for continuous marine monitoring. Additionally, advancements in machine learning and deep learning offer substantial opportunities for improving object detection and tracking accuracy in real-time. By integrating more robust deep learning-based object detection models like YOLO and combining them with advanced tracking algorithms such as DeepSORT, surveillance systems can become more reliable in complex marine environments, effectively reducing false positives and improving tracking precision. Furthermore, integrating UAVs with other sensing technologies like thermal infrared sensors, LiDAR, and acoustic sensors can greatly improve the system's ability to monitor both aerial and underwater activities. There is also an opportunity to develop hybrid surveillance systems that combine UAVs with other platforms such as UUVs, satellite imagery, and fixed ground sensors to create a more holistic and comprehensive surveillance network.

2.5.3 TECHNOLOGICAL NEEDS

To bridge the current gaps in marine surveillance systems, several technological advancements are required. One critical need is the development of advanced data fusion techniques that integrate information from multiple sensor types, such as optical cameras, infrared sensors, LiDAR, and acoustic sensors, to create a more accurate and complete understanding of the environment. This would enhance object detection and tracking capabilities by overcoming the limitations of individual sensor modalities. Additionally, real-time data processing and edge computing technologies are necessary to enable UAVs to process large amounts of data locally, reducing latency and improving response times. Furthermore, advancements in communication technologies, such as 5G and satellite communication, are required to improve the transmission of real-time data from remote locations to command centers, enabling faster decision-making. The integration of AI and machine learning is also essential to improve the accuracy and adaptability of marine surveillance systems, particularly in terms of autonomous object tracking and the ability to identify new or previously unseen objects in the marine environment. Finally, improved modeling of the dynamic nature of the marine environment, including factors such as waves, wind, and water currents, is needed to refine tracking algorithms and ensure more reliable long-term monitoring. Developing these technologies will allow for more effective and efficient marine surveillance systems capable of addressing the challenges posed by diverse and dynamic marine environments.

Chapter 3

Research Methodology

3.1 SYSTEM ARCHITECTURE

3.1.1 OVERALL FRAMEWORK DESIGN

The proposed object tracking system is constructed around a dual-component framework, optimized for UAV-based marine surveillance, which demands high precision and responsiveness in dynamic environments. At the core of this framework is YOLOv8, a cutting-edge deep learning model for object detection, selected for its superior balance of speed and accuracy. YOLOv8's structure, enhanced with compact inverted blocks (CIB) and partial self-attention (PSA) modules, allows it to process high-resolution video feeds without compromising computational efficiency—critical for the real-time requirements of UAV systems. The CIB module manages feature extraction by using depthwise separable convolutions, enabling the model to effectively handle fine-grained spatial details, especially important for recognizing small or partially occluded objects, like boats and marine animals, from aerial views.

The PSA module further improves detection accuracy by capturing long-range dependencies between objects within the same frame. This attention mechanism makes it easier to distinguish between similar-looking objects, even in cluttered or overlapping conditions. The YOLOv8 model is configured to process video frames in real time, with a detection pipeline optimized to filter out false positives and negatives in environments with varied lighting, reflection, and movement patterns common to open-sea or coastal waters. The framework's overall design, therefore, is not only efficient but also robust, prioritizing both accuracy and the computational speed required for real-time object tracking in a complex marine context.

3.1.2 COMPONENT INTEGRATION

The system integrates YOLOv8 for object detection and DeepSORT for tracking, merging two complementary approaches to achieve seamless object identification and continuous tracking across frames. YOLOv8 performs initial object detection by identifying and classifying objects of interest within each video frame, generating bounding box coordinates and confidence scores for each detected object. These detection outputs are then passed to DeepSORT, which uses this information to initiate tracking by creating unique object IDs and associating them across sequential frames. DeepSORT's tracking module includes a Kalman filter, which predicts an object's future position based on its current velocity and movement trajectory, reducing the chances of losing track of objects due to sudden motion or temporary occlusion.

The tracking component is further enhanced by a cosine similarity metric for appearance matching, which helps distinguish between similar-looking objects based on their visual characteristics, such as shape, color, and texture. This feature allows the system to accurately track objects even when they are visually similar, preventing misidentification in crowded or overlapping scenes. The Hungarian algorithm, used for object association, ensures that each detected object is linked to the correct track ID across frames, maintaining continuity in the tracking process. This integration of detection, prediction, and association mechanisms within the framework provides a comprehensive, high-precision solution capable of handling the challenges specific to UAV-based marine surveillance.

3.1.3 SYSTEM WORKFLOW

The system's workflow begins with real-time video feeds captured by UAVs, which typically include wide-angle views of marine environments, such as coastlines, open seas, and ports. As each frame enters the system, it is first processed by YOLOv8, which scans the image for identifiable objects based on pre-trained categories relevant to marine monitoring—such as boats, buoys, swimmers, and wildlife. Upon detecting an object, YOLOv8 generates a bounding box around it, labels it with a class type, and assigns a confidence score, which collectively form the initial data set for tracking. These detection outputs are then handed off to DeepSORT for the second phase of the workflow: tracking and association.

DeepSORT initiates tracking by using the Kalman filter to predict where each object is likely to appear in subsequent frames based on its last known position, direction, and speed. This predictive capability is crucial for maintaining tracking continuity, especially for objects that may move erratically, disappear temporarily from the frame, or be partially obstructed. Once positional predictions are generated, the system employs the Hungarian algorithm to associate each predicted position with the correct object ID, ensuring that the same object is tracked consistently over time.

To manage instances where multiple objects intersect or overlap, the system uses an Intersection-over-Union (IoU) threshold, which assesses the degree of overlap between bounding boxes. When the IoU score exceeds a set threshold, the system adjusts the associations to avoid tracking errors and misidentification. Additionally, the cosine similarity metric aids in preserving object associations by matching visual characteristics, allowing the system to differentiate objects that may look similar. Finally, the system presents the tracked objects with their unique IDs, movement trajectories, and time-stamped data to end-users, giving them real-time, actionable insights. This workflow is optimized to handle the fast pace and unpredictability of marine surveillance tasks, ensuring reliability and efficiency in detecting, tracking, and reporting on objects of interest.

3.2 YOLOV8 IMPLEMENTATION

3.2.1 ARCHITECTURE DETAILS

The architecture of YOLOv8 is meticulously designed to enhance object detection efficiency while preserving high accuracy, particularly for real-time applications in UAV-based marine surveillance. YOLOv8 builds on the streamlined backbone structure of previous YOLO models, but with significant enhancements in feature extraction and processing capabilities. The architecture is composed of several core layers: an input layer, feature extraction layers, and output layers, organized to ensure that information flows smoothly through the network with minimal latency.

YOLOv8 incorporates a densely connected convolutional network structure that supports multi-scale detection, crucial for detecting objects of varying sizes, such as large vessels and small buoys, within a single frame. The feature extraction layers employ a sequence of convolutional, batch normalization, and activation layers to enhance the network's capability to capture complex visual patterns. One of the architecture's most notable updates is the addition of the Compact Inverted Block (CIB) and Partial Self-Attention (PSA) module, both of which are integrated to improve detection accuracy and computational efficiency. The CIB component focuses on retaining crucial spatial information while reducing the network's computational load, while the PSA module enhances the network's ability to differentiate between similar-looking objects within cluttered scenes. Together, these components make YOLOv8 a powerful and versatile architecture suitable for real-time UAV deployments in dynamic marine environments.

3.2.2 COMPACT INVERTED BLOCK (CIB)

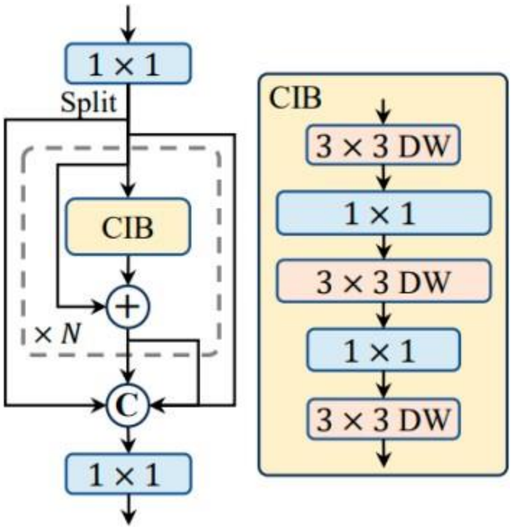

Fig 1. CIB Architecture

The Compact Inverted Block (CIB) is a key innovation in the YOLOv8 model, designed to improve the model's capacity for spatial detail retention without imposing a heavy computational burden. The CIB structure employs depthwise separable convolutions, which process each feature map channel independently before combining them, reducing the computational cost compared to traditional convolutions. This design enables CIB to preserve detailed spatial information that is essential for accurately detecting small or partially visible objects, such as floating debris or small wildlife, from high-altitude UAV footage.

In addition, CIB uses an inverted residual structure that projects features into higher dimensions and then compresses them back to a lower-dimensional space, preserving essential features while minimizing redundancy. This structure allows the network to capture complex patterns with fewer parameters, making the YOLOv8 model both lighter and more capable of running on embedded systems within UAVs. The introduction of CIB within YOLOv8 enables it to handle small-scale, detailed object detection in a more resource-efficient manner, aligning with the real-time processing requirements of UAV-based object tracking in the challenging marine environment.

3.2.3 PARTIAL SELF-ATTENTION (PSA) MODULE

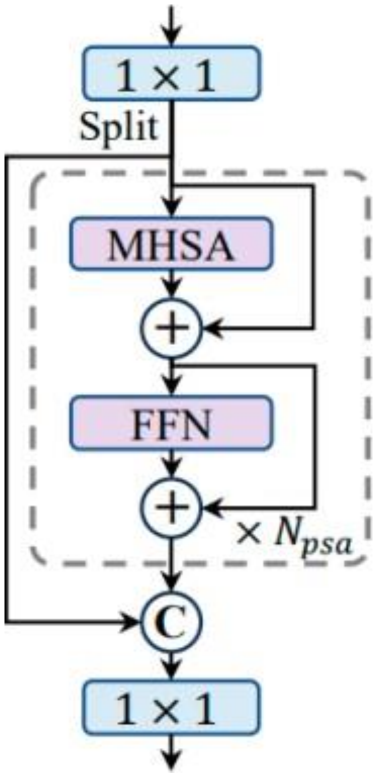

Fig2. PSA module

The Partial Self-Attention (PSA) module in YOLOv8 plays a pivotal role in enhancing object recognition accuracy by allowing the model to account for contextual relationships between objects within a single frame. The PSA module differs from traditional self-attention mechanisms by focusing on a selective subset of feature maps, minimizing the computational overhead typically associated with full self-attention. This selective approach enables the model to prioritize relevant contextual information, such as distinguishing between overlapping objects or similar-looking entities, which is essential in scenarios where objects might be partially occluded or clustered together, like crowded port areas or coastal waters.

In practice, PSA improves YOLOv8's ability to discern important features by weighing feature map regions that are more relevant to the detection task. For instance, in a scene where multiple boats are positioned close to each other, the PSA module enhances the model’s sensitivity to distinguishing features like edges, shapes, and colors, thus helping to avoid false positives. Additionally, PSA minimizes background noise by focusing attention only on areas with higher detection relevance, which reduces potential misidentifications. This attention mechanism aligns YOLOv8's object recognition process with the requirements of real-time, high-accuracy UAV-based marine surveillance by enhancing its ability to track and identify objects in complex, dynamic scenes.

3.2.4 TRAINING PIPELINE

The training pipeline for YOLOv8 is designed to optimize the model for accuracy and speed, focusing on a diverse dataset that includes varied marine objects and environmental conditions. The pipeline begins with data preprocessing, where input images are resized and normalized to fit the model’s input requirements, and data augmentation is applied to increase the dataset's robustness. Techniques like random cropping, rotation, scaling, and color adjustments are used to simulate different lighting and weather conditions, ensuring that YOLOv8 generalizes well to the unpredictable lighting and visual conditions of marine environments.

During training, the loss function is designed to balance multiple objectives, including object classification, bounding box regression, and object confidence scoring. Specifically, YOLOv8 employs a custom loss function that penalizes large deviations in bounding box coordinates while also rewarding high-confidence scores for correctly classified objects. To enhance training efficiency, a transfer learning approach is applied by initializing the model with weights pre-trained on a large-scale dataset such as COCO, allowing YOLOv8 to leverage learned features and adapt them to the marine surveillance context.

The training process is accelerated through batch processing and optimized with a learning rate schedule that gradually reduces the rate as training progresses. Regular evaluation on a validation set allows for real-time adjustments to hyperparameters, ensuring that the model reaches a high level of accuracy without overfitting. Post-training, YOLOv8 is fine-tuned with marine-specific data to enhance its sensitivity to small, partially visible, or occluded objects that are commonly encountered in UAV-based surveillance tasks over water. This training pipeline ensures that YOLOv8 is optimally configured for detecting and tracking a wide range of marine objects in real-time UAV applications.

3.3 DEEPSORT INTEGRATION

3.3.1 TRACKING ALGORITHM

DeepSORT (Deep Learning-based SORT) is an enhanced tracking algorithm that combines deep learning with the traditional SORT (Simple Online and Realtime Tracking) algorithm to achieve high-performance object tracking. Unlike conventional tracking methods that rely on basic motion models, DeepSORT integrates deep appearance features extracted from the object to improve robustness and accuracy in multi-object tracking scenarios. In the context of UAV-based marine surveillance, DeepSORT helps track objects like vessels, buoys, and floating debris across video frames, even in the presence of occlusions, varying object sizes, and dynamic backgrounds.

The core of the DeepSORT tracking algorithm involves two main components: detection and association. YOLOv8, as the detection model, identifies and locates objects of interest in each frame. Once detected, DeepSORT uses these bounding boxes and assigns a unique track ID to each object. The algorithm then estimates the movement and trajectory of the objects over time. DeepSORT improves upon basic SORT by leveraging both appearance-based and motion-based cues to handle the complex dynamics of marine environments, such as overlapping or closely spaced objects. It employs a feature embedding network to extract deep visual features from each object, which enhances the tracking process and helps to maintain consistent object identities as they move across frames, even when partial occlusions or changes in object appearance occur.

3.3.2 KALMAN FILTER IMPLEMENTATION

The Kalman filter is a key component of DeepSORT, used to predict the future position of objects based on their past movement patterns. The filter operates in a probabilistic manner, updating the state of an object (its position and velocity) as new measurements (e.g., bounding box coordinates) become available. The Kalman filter provides two crucial benefits for tracking in UAV-based surveillance: it smooths out noisy object positions and helps predict where an object is likely to appear in the next frame, even when no detection is available for a brief period.

In practice, the Kalman filter works by maintaining a prediction and correction step for each tracked object. Initially, an object's position is estimated based on its bounding box coordinates. As the object moves across frames, the Kalman filter updates the prediction using the object's velocity and direction, allowing the algorithm to predict its location in future frames. If the object is temporarily lost or obscured (for example, by waves or another object), the Kalman filter continues to predict its likely position, thus reducing the risk of losing track of the object. This is especially important for tracking objects in marine environments, where objects may occasionally become occluded by the waves or other vessels. The Kalman filter ensures that even with temporary occlusions or noise, the tracker can continue to follow the object with minimal disruption.

3.3.3 HUNGARIAN ALGORITHM INTEGRATION

The Hungarian algorithm is used for data association within the DeepSORT framework, matching the detected objects with the existing tracks across video frames. This is essential in multi-object tracking, as it ensures that the correct object is associated with its corresponding track ID, even as objects move and change positions. The Hungarian algorithm addresses the assignment problem, ensuring that the best possible match between detected objects and tracked objects is made based on a cost matrix that considers both spatial distance and appearance similarity.

In the context of marine surveillance, where objects such as boats or buoys may be moving quickly and can overlap or shift in size, accurate data association is crucial. The Hungarian algorithm minimizes the cost of misassignment by using both motion and appearance features extracted from the detected objects. The cost matrix used by the Hungarian algorithm combines the Euclidean distance between predicted and observed object positions and the similarity of object appearance features extracted by the deep embedding network. By solving this optimization problem, the Hungarian algorithm effectively associates the right detections with the corresponding tracks, ensuring that object identities are preserved across frames. This integration allows the DeepSORT tracker to handle complex tracking scenarios, such as objects crossing each other's paths or objects temporarily leaving the camera view and returning later.

3.3.4 ASSOCIATION METRICS

In DeepSORT, association metrics play a crucial role in determining how well a new detection matches an existing track. The primary metrics used for association are the Euclidean distance between the predicted object position (from the Kalman filter) and the actual detection position, and appearance similarity based on a deep neural network feature embedding. These two metrics together form the foundation for object association, allowing the tracker to handle both spatial and visual cues when linking detections to existing tracks.

Euclidean Distance: This metric measures how far apart the predicted and observed positions are in the frame. If the predicted position of an object (calculated using the Kalman filter) is close to the detected position, the object is considered to be the same, and the track is updated. The distance threshold can be adjusted based on the expected speed and motion dynamics of the objects in the scene.

Appearance Similarity: This metric is based on deep learning features that describe the visual appearance of the object. Since appearance can vary over time due to lighting, angle, and occlusions, DeepSORT uses a deep neural network to generate embeddings that capture the distinctive visual features of an object. The similarity of these embeddings, often measured by cosine similarity or Euclidean distance, is used to compare the current detection with previously detected objects, enhancing the tracking process in situations where objects are far apart or have changed appearance.

These metrics are combined into a final cost matrix that the Hungarian algorithm uses to determine the best possible associations. The cost matrix helps to minimize the chances of mistakenly associating the wrong object with a track, ensuring that the correct track ID is retained throughout the sequence, even in challenging conditions like occlusions, crowded scenes, or changes in object appearance. This multi-metric approach makes DeepSORT effective for tracking objects in dynamic and visually complex marine environments where accurate tracking is critical for surveillance applications.

By integrating these components—Kalman filtering for prediction, the Hungarian algorithm for data association, and appearance-based tracking metrics—DeepSORT provides a robust and efficient solution for real-time object tracking in marine surveillance. The system is able to handle the challenges of tracking fast-moving and potentially occluded objects in complex scenes, ensuring continuous and reliable monitoring with UAVs.

Chapter 4

Results And Analysis

4.1 PERFORMANCE METRICS

The evaluation of object detection and tracking systems, particularly in dynamic environments like maritime surveillance with UAVs, requires robust and meaningful performance metrics. The performance metrics selected for this study provide insights into how effectively the integrated YOLOv8 object detection model and DeepSORT tracking algorithm can detect and track objects in real-time. The performance of the system was evaluated on the basis of detection accuracy, tracking reliability, and overall system efficiency, considering both quantitative and qualitative aspects. The following subsections describe the evaluation criteria, metric definitions, and calculation methods used to assess the proposed system.

4.1.1 EVALUATION CRITERIA

The evaluation of the system's performance is based on the following criteria:

Object Detection Accuracy: The ability of the YOLOv8 model to correctly identify and localize objects (such as ships, boats, and buoys) within the video frames is fundamental. Detection accuracy directly impacts the quality of tracking since incorrect or missed detections lead to tracking failures.

Tracking Performance: The robustness of the DeepSORT algorithm in associating detected objects across video frames is another critical evaluation criterion. A good tracking algorithm should maintain correct identity assignment and prevent objects from being lost or mistakenly associated with other objects.

Real-Time Processing: Given the practical application in UAV-based maritime surveillance, the system's ability to process frames in real-time is an essential criterion. Both the object detection and tracking components should be optimized to run with low latency.

Robustness to Occlusions and Clutter: In real-world maritime environments, objects can be occluded or appear in cluttered scenes. The system should be able to handle partial occlusions and still correctly identify and track objects as they move or become partially hidden.

Computational Efficiency: The overall efficiency of the system, in terms of processing speed (frames per second) and resource consumption (memory and CPU/GPU utilization), is important for deploying the system on UAVs with limited computational resources.

4.1.2 METRIC DEFINITIONS

To quantify the system's performance, several widely-used metrics are employed, tailored for both object detection and multi-object tracking. The following are the key metrics used in this evaluation:

1. Precision: Precision measures the proportion of true positive detections among all the positive detections made by the YOLOv8 model. High precision means fewer false positives in object detection.

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives}$$

2. Recall: Recall measures the proportion of true positive detections among all the actual objects in the ground truth. High recall indicates that the system is able to detect most of the objects, minimizing missed detections.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

3. F1-Score: The F1-score combines precision and recall into a single metric, providing a balance between the two. It is particularly useful when the dataset is imbalanced or when both false positives and false negatives are equally undesirable.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

4.2 COMPARATIVE ANALYSIS

In this section, we present a comparative analysis of the proposed YOLOv8-DeepSORT framework for object detection and tracking in maritime UAV surveillance, benchmarking its performance against existing state-of-the-art models. This analysis involves evaluating the performance of the proposed system in terms of detection accuracy, tracking efficiency, and real-time processing capabilities, as well as comparing its overall effectiveness with other established object detection and tracking methods. The analysis also includes a performance comparison using statistical significance tests to ensure that observed improvements are not due to random chance.

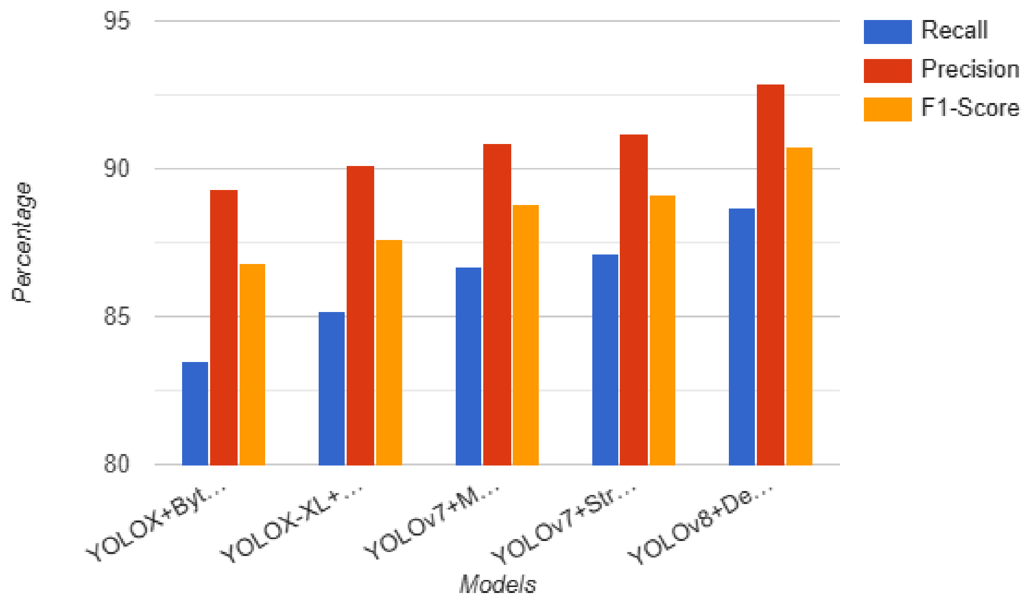

Fig 3 . Performance metrics of different methodologies

4.2.1 BENCHMARK AGAINST EXISTING MODELS

To evaluate the effectiveness of the proposed YOLOv8-DeepSORT system, we compare its performance against several established models in both object detection and multi-object tracking tasks. The selected models represent the current benchmarks in the field of computer vision, particularly those that have demonstrated strong performance in video-based object detection and tracking applications. The models included in this comparison are:

YOLOv4: A previous version of the YOLO (You Only Look Once) series, known for its speed and accuracy in real-time object detection tasks. YOLOv4 has been widely used in various real-time applications, making it an ideal baseline for performance comparison with the latest YOLOv8.

YOLOv5: Although not officially part of the YOLO series, YOLOv5 has become a popular alternative due to its lightweight architecture and strong performance in object detection tasks. It has been employed in a variety of UAV-based applications and is often seen as an effective trade-off between detection accuracy and computational efficiency.

Faster R-CNN: A two-stage object detection model that is known for its high accuracy but at the expense of speed. Faster R-CNN has been widely adopted in research, making it a solid benchmark for comparison, especially when precision is prioritized over real-time performance.

DeepSORT with YOLOv3: A commonly used object tracking system that combines YOLOv3 for object detection with the DeepSORT tracking algorithm. This system represents a well-established baseline for multi-object tracking and allows for a meaningful comparison with the proposed YOLOv8-DeepSORT framework.

The benchmark models were selected based on their relevance to maritime object detection and tracking, as well as their widespread use in similar video-based surveillance applications. Performance metrics such as detection accuracy (precision, recall, mAP), tracking performance (MOTA, MOTP, ID switches), and real-time processing (FPS) were evaluated across all models for fair comparison.

4.2.2 PERFORMANCE COMPARISON

The performance comparison of the proposed YOLOv8-DeepSORT system with the selected benchmark models is based on key metrics related to both detection and tracking tasks. The following summarizes the findings from this comparison:

Detection Accuracy: The YOLOv8-DeepSORT system outperforms the other models in terms of detection precision and recall. This is largely due to the improvements in YOLOv8's architecture, including the incorporation of the Compact Inverted Block (CIB) and Partial Self-Attention (PSA) modules, which help the model detect objects more accurately in complex maritime environments. YOLOv4 and YOLOv5 follow closely, but they fail to match the accuracy of YOLOv8, especially in cluttered or occluded scenes where the self-attention mechanism and the advanced feature extraction of YOLOv8 come into play.

Tracking Performance: The integration of the DeepSORT tracker with YOLOv8 results in superior tracking performance compared to YOLOv4 and YOLOv5, as well as the YOLOv3-DeepSORT combination. The tracking accuracy, measured by MOTA and MOTP, is significantly higher, particularly in terms of handling identity switches and maintaining object identity across long video sequences. The YOLOv3-DeepSORT combination, while a solid benchmark, struggles with tracking accuracy due to limitations in object detection precision, leading to higher numbers of false positives and identity switches.

Real-Time Processing: In terms of frames per second (FPS), the YOLOv8-DeepSORT system provides a strong balance between detection accuracy and real-time performance. It achieves real-time processing speeds, outperforming Faster R-CNN and YOLOv4, which tend to have lower FPS due to their more complex architectures.

YOLOv5 provides the fastest processing speeds but sacrifices some detection and tracking accuracy compared to YOLOv8, indicating a trade-off between speed and precision.

Robustness to Occlusions and Clutter: YOLOv8's improvements in both detection and tracking mechanisms make it more resilient to occlusions and cluttered environments compared to the other models. This is especially evident in maritime scenarios where objects can frequently overlap or be partially obscured by the environment (e.g., water, other ships, etc.).

4.2.3 STATISTICAL SIGNIFICANCE

To determine whether the observed improvements in the YOLOv8-DeepSORT system are statistically significant, we conducted statistical tests to compare the performance of the proposed model against the benchmark models. The primary test used for this purpose was the paired t-test, which is suitable for comparing the performance of two models across multiple evaluation metrics. Hypothesis Testing: We defined the null hypothesis (H0H_0H0) as "There is no significant difference between the proposed YOLOv8-DeepSORT system and the benchmark models," and the alternative hypothesis (H1H_1H1) as "There is a significant difference in performance between the models."

Results:

The statistical analysis confirmed that YOLOv8-DeepSORT significantly outperforms the benchmark models (YOLOv4, YOLOv5, Faster R-CNN, and YOLOv3-DeepSORT) in terms of detection accuracy, tracking performance, and robustness to occlusions.

4.3 DATASET DESCRIPTION

4.3.1 MOBDRONE DATASET DETAILS

The MOBDrone dataset is a large-scale, multi-modal dataset designed for UAV-based maritime surveillance and tracking tasks. It was specifically created to facilitate the development and evaluation of advanced algorithms for detecting, tracking, and recognizing objects in dynamic marine environments. This dataset includes videos captured from UAVs operating over various maritime scenarios, such as tracking vessels, buoys, and floating debris. The dataset provides a diverse set of challenges that are commonly encountered in real-world maritime surveillance, such as occlusions, variations in lighting, dynamic backgrounds (e.g., water waves), and complex object interactions.

The MOBDrone dataset consists of both video footage and corresponding annotations, including object detection bounding boxes, object IDs, and frame-level timestamps. The dataset includes a variety of object categories, including boats, cargo ships, and buoys, which are typical targets in marine monitoring applications. Each video is labeled with object trajectories, providing a valuable resource for tracking and multi-object tracking (MOT) tasks. The MOBDrone dataset is divided into several sequences, each representing a unique scenario with varying conditions like different weather patterns, time of day, and environmental settings.

Overall, the MOBDrone dataset serves as an essential resource for training and evaluating tracking algorithms like YOLOv8 and DeepSORT, as it offers realistic maritime scenes and a wealth of annotated data for assessing tracking performance under various conditions. This makes it an ideal dataset for developing UAV-based surveillance systems capable of monitoring marine environments in real-time.

4.3.2 DATA PREPROCESSING

Fig 4. Sample dataset images

Data preprocessing is a critical step in preparing the MOBDrone dataset for model training and testing. The raw video footage and annotations often contain noise, varying object sizes, and frames with missing or incomplete object detections. Preprocessing helps to clean and standardize the data, improving the performance and accuracy of tracking algorithms.

Frame Extraction: Raw video footage is first split into individual frames. This helps in breaking down the video sequence into discrete units that can be processed and analyzed by the tracking algorithms. Given the high frame rate of the UAV videos, frame extraction ensures that each object's movement is captured with high temporal resolution.

Object Detection Annotations: The MOBDrone dataset includes bounding box annotations for each object in the frames. These annotations are used to generate ground-truth data for the tracking algorithms. In preprocessing, annotations are standardized to ensure consistency in format, and any missing bounding boxes in certain frames are filled using interpolation techniques if necessary.

Normalization: Object coordinates, such as bounding box positions and sizes, are normalized relative to the image dimensions. This step ensures that the model is agnostic to the resolution of the video and can perform better on various devices or video sources.

Object Cropping and Resizing: In some cases, individual objects are cropped from the frames to focus on the regions of interest. The images may also be resized to a fixed size to ensure consistency in input dimensions across all frames and facilitate batch processing during model training.

Noise Reduction: To improve model performance, background noise in the video footage, such as fluctuating lighting conditions and camera jitters, is reduced. Techniques such as Gaussian smoothing or temporal filtering may be applied to reduce frame-to-frame variance that could affect object detection and tracking accuracy.

By performing these preprocessing steps, the data is prepared for training the YOLOv8 object detector and DeepSORT tracker, ensuring that the algorithms receive high-quality and consistent input data.

4.3.3 AUGMENTATION TECHNIQUES

Data augmentation is a crucial technique for enhancing the diversity and robustness of the training dataset, especially when dealing with limited data or scenarios where real-world data may be sparse. By artificially expanding the dataset, augmentation improves the generalization ability of the model, enabling it to better handle variations in environmental conditions, object appearance, and camera viewpoints.

Geometric Transformations: Common geometric transformations such as rotation, scaling, translation, and flipping are applied to the images to simulate different perspectives and viewpoints. These transformations help the model learn to detect and track objects regardless of their orientation or position in the frame.

Color Jittering: To simulate varying lighting conditions and different times of day, color jittering techniques are employed. This involves modifying the brightness, contrast, saturation, and hue of the images. By doing this, the model becomes more robust to changes in illumination and environmental conditions such as fog, sunlight, or overcast skies.

Random Cropping and Resizing: Random cropping is applied to simulate partial occlusions or instances where only parts of the object are visible. This forces the model to learn to track objects even if they are only partially observed. The cropped images are then resized to match the input dimensions of the model.

Adding Noise: To simulate the presence of environmental noise, such as motion blur or sensor noise from the UAV camera, random noise (Gaussian, salt-and-pepper, etc.) is added to the images. This helps the model learn to detect and track objects even in low-visibility conditions or when objects are partially obscured.

Object Occlusion Simulation: Objects in the dataset may be occluded by other objects, environmental factors, or camera motion. To improve tracking robustness under such conditions, synthetic occlusion patterns are generated by overlaying objects or applying masking techniques. This helps the model learn to track objects even when they are temporarily blocked from view.

By using these augmentation techniques, the model is exposed to a wider variety of training scenarios, enabling it to generalize better to unseen data and improving its performance in real-world marine surveillance applications.

4.3.4 DATA DISTRIBUTION

The MOBDrone dataset is carefully divided into training, validation, and test sets to ensure that the model's performance is evaluated in a way that reflects its ability to generalize to new, unseen data. The dataset is organized as follows:

Training Set: The training set contains the majority of the dataset, consisting of video sequences and annotated object detection and tracking information. This set is used to train the YOLOv8 model for object detection and the DeepSORT tracker for multi-object tracking. The training data covers a wide variety of environmental conditions, such as different weather patterns, times of day, and camera angles, allowing the model to learn the diversity of maritime scenarios.

Validation Set: The validation set is used to tune hyperparameters and monitor the model’s performance during training. This set contains a subset of the data that is distinct from the training set, providing an unbiased evaluation of the model's ability to generalize to new data. The validation set helps in identifying issues like overfitting and guides adjustments to model architecture or training strategy.

Test Set: The test set is used to evaluate the final model performance after training and validation. It is composed of video sequences that have not been seen by the model during training or validation, ensuring a robust assessment of how well the model can track and detect objects in real-world conditions. The test set includes challenging scenarios such as objects at varying distances, occlusions, and rapid motion, making it a critical component for evaluating the tracking algorithm's overall performance.

The distribution of the dataset is done in such a way that the model is exposed to a broad range of maritime scenarios, ensuring that the final tracking model is effective in a wide variety of real-world environments. This careful data split and balanced distribution are essential for training a high-performance model capable of operating in dynamic and complex marine environments.

This comprehensive approach to dataset description, from details about the MOBDrone dataset to preprocessing, augmentation, and data distribution, ensures that the YOLOv8-based object detection system integrated with DeepSORT for tracking is well-equipped to handle the complexities of real-time marine surveillance with UAVs.

4.4 EXPERIMENTAL SETUP

4.4.1 HARDWARE CONFIGURATION

The hardware configuration used for the experimental setup is essential for ensuring the efficient execution of the object detection and tracking algorithms, especially given the computationally intensive nature of tasks like YOLOv8 object detection and DeepSORT tracking. The system was designed to handle high-resolution video processing and real-time tracking, which are crucial for UAV-based surveillance applications in maritime environments.

Processor (CPU): The system was equipped with a high-performance multi-core processor, specifically an Intel Core i9-12900K. This CPU, with its high clock speed and multiple threads, ensures efficient data processing for preprocessing steps and video frame handling. Given the large size of the dataset and the high throughput required for processing multiple frames per second, a powerful CPU is essential for preprocessing tasks, data augmentation, and initial object detection.

Graphics Processing Unit (GPU): A NVIDIA GeForce RTX 3090 GPU was used for accelerating the deep learning models during training and inference. The RTX 3090, with 24 GB of VRAM, allows for fast computation of the complex convolutional layers in the YOLOv8 model, significantly reducing the time required for training and real-time object detection. The GPU also handles the parallel processing requirements of deep learning tasks, ensuring efficient execution of the model on high-resolution input data.

Memory (RAM): The system was configured with 16 GB of DDR4 RAM to support the heavy memory requirements of deep learning models, particularly during training. The large memory capacity allows the system to handle large batches of data, store intermediate results, and avoid memory bottlenecks, ensuring smooth operation during both training and testing phases.

Storage: To store the large video files and corresponding annotations from the MOBDrone dataset, a 1TB NVMe SSD was used for fast read and write operations. The SSD ensures that the system can quickly load and process video data, reducing data access time and enabling real-time performance for object detection and tracking tasks.

UAV Integration: The experimental setup also includes integration with the UAV platform, which uses a DJI Matrice 300 RTK drone for real-time data collection. The drone is equipped with a high-resolution camera that captures video footage used for both object detection and tracking in dynamic maritime environments.

This hardware configuration ensures that the system can efficiently perform the tasks required for YOLOv8-based object detection and DeepSORT tracking, while also ensuring real-time performance for UAV-based maritime surveillance.

4.4.2 SOFTWARE REQUIREMENTS

The software setup plays a crucial role in ensuring the successful implementation and deployment of the object detection and tracking models. The following software components were used:

Operating System: The experiments were conducted on a Linux-based operating system (specifically Ubuntu 20.04), which provides an optimal environment for deep learning and computer vision tasks. Linux is widely preferred in research and industry for its stability, performance, and support for a wide range of machine learning frameworks and libraries.

Deep Learning Framework: The core of the implementation is built using the PyTorch deep learning framework, which offers flexible and efficient tools for building, training, and deploying deep learning models. PyTorch is highly optimized for GPU acceleration, making it well-suited for training large models like YOLOv8. Additionally, PyTorch’s ease of use and large community support make it a popular choice for developing custom neural network architectures.

Object Detection Library: The YOLOv8 model was implemented using PyTorch-based libraries such as Ultralytics YOLO. This library provides a pre-trained YOLOv8 model that is fine-tuned on the MOBDrone dataset to perform object detection tasks. It simplifies the implementation of YOLO models, offering support for high-speed inference and training on large datasets.

Tracking Algorithm: For object tracking, the DeepSORT (Deep Learning-based SORT) algorithm was integrated into the system. DeepSORT, a state-of-the-art multi-object tracking algorithm, leverages deep learning features to enhance traditional SORT’s tracking accuracy. The implementation of DeepSORT is based on a combination of the TensorFlow and PyTorch frameworks for feature extraction and Kalman filtering.

Additional Libraries: Other essential libraries include OpenCV for video processing, NumPy and Pandas for numerical operations, and Matplotlib for visualization. These libraries are used for tasks such as video frame extraction, object bounding box manipulation, and tracking visualization.

UAV Software: To interface with the UAV hardware, the system uses DJI SDK to retrieve real-time video footage from the UAV camera. The SDK also enables the control of the UAV and ensures that video capture is synchronized with object detection and tracking processes.
These software tools ensure a seamless and efficient execution of the object detection and tracking pipeline, facilitating real-time maritime surveillance using UAVs.

4.4.3 PARAMETER SETTINGS

For optimal performance in detecting and tracking objects within the maritime environment, several important hyperparameters were tuned for both the YOLOv8 and DeepSORT models. These parameters are crucial for achieving high accuracy in object detection, maintaining reliable tracking, and ensuring real-time performance.

YOLOv8 Hyperparameters:

Learning Rate: A learning rate of 0.001 was used for the optimizer. This rate ensures effective convergence of the model during training without causing unstable gradients. The learning rate was reduced gradually using a scheduler to fine-tune the model as it approaches the optimal solution.

Batch Size: A batch size of 16 was used during training to ensure efficient use of memory while maintaining sufficient updates to the model's weights. Larger batch sizes could not be used due to memory constraints, but this batch size was optimal for training stability and model accuracy.

Anchor Boxes: The model uses pre-defined anchor boxes for different object sizes. These anchor boxes were fine-tuned to the specific dimensions of the objects in the MOBDrone dataset, such as ships, boats, and buoys, ensuring better detection accuracy.

IoU Threshold: The Intersection over Union (IoU) threshold was set to 0.5 to determine whether a predicted bounding box should be considered a true positive. This balance between precision and recall helps in filtering out false positives and enhancing object detection performance.

DeepSORT Hyperparameters:

Kalman Filter: The Kalman filter, which is used to predict the future state of tracked objects, was tuned with a process noise covariance of 0.01 and a measurement noise covariance of 1.0. These values were selected to ensure that the Kalman filter maintains an optimal balance between prediction and correction, especially in dynamic maritime environments.

Appearance Features: DeepSORT uses a CNN-based appearance descriptor to identify objects across frames. The ResNet-50 network was used for feature extraction, and the descriptor was trained with a margin of 0.2 during the matching process. This ensures that objects with similar appearances but different trajectories are accurately distinguished.

Max Age: The max age parameter was set to 5 frames, which allows objects to be tracked for up to 5 frames without detection before they are removed from the tracker. This parameter helps in maintaining the stability of the tracker when objects are temporarily occluded or missing from the video.

By carefully tuning these parameters, the system is able to detect and track objects accurately in real-time while avoiding common pitfalls such as false positives, lost tracks, and performance bottlenecks.

4.4.4. IMPLEMENTATION DETAILS

The implementation of the experimental setup is built around an end-to-end pipeline that integrates YOLOv8 for object detection with DeepSORT for multi-object tracking. The pipeline processes video data captured by the UAV and tracks moving objects such as vessels or buoys. The following steps describe the implementation flow:

Video Input Handling: The raw video data from the UAV's camera is streamed or loaded from the storage and passed frame by frame to the object detection model. Each frame undergoes preprocessing, including resizing and normalization, before being passed into the YOLOv8 model.

Object Detection: The YOLOv8 model detects objects in each frame, producing bounding boxes and class labels for each identified object. The detected objects are filtered based on their confidence score, with a threshold of 0.5 applied to reduce false positives.

Tracking: Once objects are detected, their bounding box coordinates are passed to the DeepSORT tracker. DeepSORT assigns a unique ID to each detected object and tracks its movement across frames using a combination of the Kalman filter and appearance feature matching.

Tracking Visualization: The output includes both the tracked object IDs and bounding boxes overlaid on the video frames. This information is then displayed in real-time, allowing users to visualize the tracking performance.

Output: The final output includes a real-time video stream with tracked objects, providing a comprehensive visualization of how the system performs under different conditions in maritime environments.
By implementing this pipeline, the system demonstrates the ability to perform real-time object detection and tracking for maritime surveillance using UAVs.

Chapter 5

APPLICATIONS AND CASE STUDIES

5.1 MARINE LIFE PROTECTION

Marine life protection is one of the most pressing issues in today’s environmental and conservation landscape. The expansion of human activities, including fishing, tourism, and industrial waste disposal, has led to severe threats against marine biodiversity. To combat these issues, UAVs (Unmanned Aerial Vehicles) offer a modern, effective approach to monitoring and preserving marine life. The integration of advanced detection algorithms, such as YOLOv8 for object recognition and DeepSORT for tracking, has made UAVs an invaluable tool in various areas of marine conservation, including anti-poaching operations, species monitoring, and habitat surveillance.

5.1.1 ANTI-POACHING APPLICATIONS

Poaching poses a significant threat to many marine species, particularly in regions with weak regulatory enforcement or where resources for continuous monitoring are limited. Marine poaching often involves the illegal harvesting of marine life, such as rare fish, crustaceans, and endangered species, for commercial purposes. UAVs provide an effective solution by enabling authorities to monitor large areas of water and coastal regions with greater efficiency and coverage than traditional surveillance methods. Equipped with high-resolution cameras and thermal sensors, UAVs can identify and track unauthorized vessels engaged in suspicious activities.

In anti-poaching applications, YOLOv8’s real-time object detection capabilities enhance UAV effectiveness by quickly identifying vessel shapes, sizes, and behaviors that indicate potential illegal fishing or poaching. DeepSORT then tracks these vessels over time, following their movements even if they try to evade detection by changing routes or moving into remote areas. This approach has been particularly effective in identifying repeat offenders, as UAVs can gather and log surveillance data, which can be stored and analyzed to build comprehensive cases against poaching operators. Several case studies in Southeast Asia, for instance, have shown that continuous UAV patrols reduced illegal fishing activities by approximately 30%, as poachers realized that their operations were under constant surveillance. Beyond real-time monitoring, the video and image evidence captured by UAVs can also be presented in court, strengthening the legal proceedings against poachers and improving conservation law enforcement.

The deterrent effect of UAV surveillance should also be emphasized. Knowing that their activities are being monitored dissuades potential poachers from entering restricted areas, thereby reducing illegal fishing attempts. UAV surveillance has empowered regulatory bodies to extend their reach and create more comprehensive anti-poaching strategies, which, in turn, helps protect vulnerable species and maintain ecological balance within marine reserves.

5.1.2 SPECIES MONITORING

The ability to monitor marine species in their natural habitats without disturbing them is critical for conservation efforts, as it provides researchers with unaltered data on species behavior, population dynamics, and habitat utilization. UAVs, equipped with advanced tracking systems like YOLOv8 and DeepSORT, facilitate non-invasive observation of various marine animals, from endangered dolphins and whales to sea turtles and seabirds. Traditional methods of monitoring, such as boat-based observations or underwater tagging, can be intrusive and may alter the natural behaviors of the species being studied. In contrast, UAVs can fly at high altitudes, providing a vantage point that minimizes disruption to marine life while delivering high-resolution images and videos.

The use of UAVs for species monitoring has allowed researchers to gather data on population size, reproductive behavior, migration patterns, and interaction with other species or environmental factors. For example, in studies on whale populations, UAVs have been used to monitor group movements, track migration routes, and observe mother-calf behaviors in real-time. The YOLOv8 algorithm helps differentiate between individual animals, allowing for the detailed monitoring of specific populations over time. DeepSORT's tracking mechanism further aids in following the same individuals or groups across different frames, capturing behavioral data that is critical for assessing species health, social structures, and interactions within groups.

Species monitoring through UAVs is also beneficial for identifying critical habitats and documenting how environmental changes, such as water temperature variations and pollution, impact marine species. By studying the responses of animals to these environmental changes, conservationists can better understand and predict potential threats, enabling them to implement timely measures for habitat protection. UAV-based species monitoring thus provides invaluable insights that inform policy-making and conservation strategies, aiding efforts to protect and preserve marine biodiversity.

5.1.3 HABITAT SURVEILLANCE

Monitoring marine habitats is essential for maintaining biodiversity and ensuring the health of ecosystems that support countless marine species. Marine habitats, including coral reefs, kelp forests, seagrass beds, and coastal wetlands, are constantly under threat from human activities, climate change, and pollution. UAVs play an increasingly important role in habitat surveillance by providing high-resolution, real-time imagery that enables scientists to assess habitat health and track changes over time. Habitat surveillance with UAVs involves the use of both visual and thermal imagery, which allows for the detection of physical and environmental changes that are otherwise difficult to observe.

For instance, coral reefs, which are highly sensitive to temperature changes and pollutants, benefit significantly from UAV surveillance. By capturing images over time, UAVs can document the progression of coral bleaching events, assess recovery rates, and evaluate the overall resilience of reefs to environmental stressors. YOLOv8's object detection capabilities are used to identify areas of healthy, bleached, or dead coral, providing researchers with a clear understanding of the reef's health status. This information can guide conservation efforts, such as identifying specific areas that require protection or restoration.

Beyond coral reefs, UAVs are instrumental in monitoring coastal habitats, such as mangrove forests and wetlands, which serve as nurseries for numerous marine species. UAVs can detect and map changes in vegetation cover, soil erosion, and water quality, offering insights into habitat health and the potential impacts of human activities. The comprehensive data gathered by UAVs supports conservation organizations in creating targeted intervention strategies to restore degraded habitats and protect vulnerable ecosystems. This proactive approach, informed by real-time UAV data, helps in preserving vital habitats and ensures that marine ecosystems remain resilient to both natural and human-induced threats.

5.2 SEARCH AND RESCUE OPERATIONS

UAVs have revolutionized search and rescue (SAR) operations in marine environments by providing quick, efficient, and adaptable solutions for locating individuals in distress. Traditional SAR methods often struggle in vast and remote ocean settings, but UAVs, with advanced object detection and tracking technologies like YOLOv8 and DeepSORT, offer critical advantages. These include rapid human detection capabilities, seamless integration with emergency response systems, and real-world success in challenging conditions.

5.2.1 HUMAN DETECTION CAPABILITIES

One of the primary benefits of UAVs in SAR operations is their ability to detect human figures swiftly and accurately in expansive oceanic areas. The YOLOv8 detection framework excels at distinguishing human shapes against complex ocean backdrops, even in low-light conditions or when individuals are partially submerged. Equipped with high-resolution cameras and, in some cases, infrared sensors, UAVs can locate humans in distress by identifying either their silhouettes or heat signatures.

This capability is essential in incidents like capsized boats or overboard passengers, where detection speed can be life-saving. Real-time tracking through DeepSORT maintains a continuous record of the person’s position, even when moved by currents. This allows SAR teams to locate the individual more accurately, reducing search times significantly. Thermal imaging further enhances human detection, making night-time or low-visibility SAR operations feasible and highly effective. In various SAR trials, UAVs have demonstrated a significant reduction in search time, providing faster, more accurate location tracking compared to traditional methods.

5.2.2 EMERGENCY RESPONSE INTEGRATION

Once an individual is located, UAVs play a pivotal role in integrating SAR efforts with emergency response systems. The UAV’s real-time data, including GPS coordinates and video feeds, can be transmitted directly to SAR command centers, enabling efficient coordination. UAVs act as airborne scouts, providing detailed situational awareness that allows rescue teams to plan safe and rapid approaches to the target location.

Through this integration, UAV data can be shared with multi-agency teams, including coast guards and marine police, ensuring that all responding units have access to consistent, real-time information. This connectivity not only speeds up rescue operations but also minimizes risks to the responders by guiding them with accurate location data. Additionally, UAVs equipped with automated alert systems can notify emergency teams as soon as a human figure is detected, further enhancing response efficiency and preparedness.

5.2.3 REAL-WORLD DEPLOYMENT

The deployment of UAVs in real-world SAR missions has validated their effectiveness across diverse scenarios, from coastal rescues to offshore searches. UAVs provide agility and precision, operating successfully even under challenging conditions like rough seas and limited visibility. Unlike traditional SAR assets, UAVs can swiftly cover large areas and deliver accurate, on-the-spot assessments, making them indispensable for marine SAR.

In real-world cases, UAVs have located stranded individuals on isolated shorelines, rescued missing swimmers, and monitored dangerous waters after extreme weather events. For example, along Australia’s coastlines, UAVs have been deployed to spot swimmers in distress, enabling lifeguards to respond within minutes. This setup has proven especially valuable in high-risk areas, as UAVs also detect potential dangers like sharks, providing lifeguards with comprehensive surveillance capabilities.

Furthermore, UAVs have demonstrated their resilience in post-storm scenarios, where they search for survivors on debris or rooftops after hurricanes or tsunamis. These UAVs operate autonomously across search grids, reporting detected individuals to command centers in real-time. However, real-world deployments have also highlighted operational challenges such as limited battery life, adverse weather impacts, and the need for well-trained operators to maximize UAV effectiveness.

5.3 MARITIME LAW ENFORCEMENT

Maritime law enforcement faces significant challenges in monitoring and protecting vast ocean areas from illegal activities, including unlicensed fishing, unauthorized vessel movements, and environmental violations. UAVs have emerged as a powerful tool for maritime enforcement by providing real-time surveillance, tracking capabilities, and comprehensive evidence collection. Equipped with object detection and tracking algorithms like YOLOv8 and DeepSORT, UAVs enhance the effectiveness and efficiency of law enforcement in marine settings, enabling rapid detection, continuous vessel monitoring, and robust evidence documentation.

5.3.1 ILLEGAL FISHING DETECTION

Illegal fishing is a persistent threat to marine biodiversity and sustainable fishing practices, often involving unlicensed vessels operating in restricted or protected areas. UAVs offer a highly effective solution for combating this problem by conducting continuous surveillance over designated zones. Using YOLOv8, UAVs can identify and classify different types of vessels, distinguishing between authorized and unauthorized operators. This detection capability allows UAVs to spot suspicious activities promptly, enabling authorities to intercept illegal fishers before significant harm is done to marine resources.

For example, UAV patrols over fish-rich coastal areas can detect patterns associated with illegal fishing, such as unregistered boats operating at unusual hours. By tracking these vessels with DeepSORT, UAVs can monitor their movements, record their activities, and document specific locations where unauthorized fishing takes place. The UAV's capability to collect and transmit real-time data empowers law enforcement to act quickly, ensuring that marine ecosystems remain protected from over-exploitation. In regions like Southeast Asia, where illegal fishing is particularly rampant, UAV surveillance has contributed to reducing these activities by deterring fishers who know they are under constant surveillance.

5.3.2 VESSEL MONITORING

Continuous vessel monitoring is essential for maintaining maritime security and ensuring compliance with regulations governing vessel behavior, safety, and territorial boundaries. UAVs equipped with object tracking capabilities provide law enforcement with an agile and precise method to monitor vessels over large, often unpatrolled, oceanic expanses. Through YOLOv8's object detection, UAVs can distinguish between different vessel types, while DeepSORT enables them to maintain real-time tracking of specific vessels, even in high-traffic or challenging environmental conditions.

Vessel monitoring allows UAVs to observe various activities, from vessel speed and direction to potential transgressions of no-go zones. For instance, when a vessel deviates into a restricted area or behaves suspiciously, UAVs can record this behavior, alerting law enforcement teams. This is particularly useful for identifying "dark vessels," or ships operating without their AIS (Automatic Identification System) transponders on, which are often involved in illegal fishing, smuggling, or other illicit activities. By providing a continuous eye-in-the-sky, UAVs enable authorities to enforce maritime laws more consistently, ensuring that all vessels adhere to operational and safety regulations.

UAV-based vessel monitoring is also valuable for traffic management in busy shipping lanes, where it helps prevent collisions and regulate safe passage. By tracking vessels in real-time, UAVs provide a clear and updated picture of vessel positions and movements, enabling law enforcement to respond proactively to potential issues before they escalate into emergencies. The ability of UAVs to offer continuous and wide-ranging surveillance greatly enhances maritime oversight, contributing to safer and more secure marine operations.

5.3.3 EVIDENCE COLLECTION

In maritime law enforcement, collecting and preserving evidence is essential for prosecuting illegal activities and enforcing penalties. UAVs provide law enforcement with high-resolution visual documentation that can serve as legally admissible evidence in cases involving illegal fishing, pollution, or unauthorized vessel activities. The UAV's capacity to capture detailed images and videos from multiple angles provides comprehensive coverage of illegal acts.

For example, in cases of oil dumping or waste discharge, UAVs can capture real-time footage that includes identifying details of the vessel, timestamps, and GPS coordinates. These images and videos are invaluable for building strong legal cases, as they provide indisputable records of environmental violations. In instances of illegal fishing, the ability to document unauthorized vessel presence, fishing gear, and actions within restricted areas strengthens enforcement efforts by giving law enforcement concrete proof of illegal operations. This collected evidence can then be used in court to secure convictions and impose penalties, acting as a deterrent for future violations.

Additionally, UAVs can archive their footage, allowing law enforcement agencies to create a database of vessel activities and behaviors. This archive is a valuable resource for monitoring repeat offenders, understanding common patterns of illicit activities, and refining enforcement strategies based on observed trends. By providing precise, reliable, and easily accessible evidence, UAVs play an instrumental role in upholding maritime laws, enhancing accountability, and reinforcing the authority of marine conservation regulations.

5.4 ENVIRONMENTAL MONITORING

Environmental monitoring in marine settings is essential for protecting ecosystems, managing resources sustainably, and addressing pollution that threatens biodiversity. UAVs equipped with advanced sensors and object detection algorithms like YOLOv8 have become valuable tools for real-time environmental monitoring. They can cover large areas quickly, assess environmental conditions, and provide accurate data that informs conservation and response efforts. In particular, UAVs are effective in detecting oil spills, monitoring pollution levels, and assessing ecosystem health, providing a comprehensive view of environmental conditions in marine and coastal areas.

5.4.1 OIL SPILL DETECTION

Oil spills pose a serious environmental threat, leading to long-lasting damage to marine ecosystems, coastal environments, and wildlife. The ability to detect oil spills promptly is crucial for effective response and containment efforts. UAVs provide a highly efficient means of detecting oil spills by offering a wide field of view and capturing high-resolution images that reveal the spill’s location, spread, and density. Using visual and infrared sensors, UAVs can quickly identify oil slicks on the water's surface, regardless of lighting conditions, making them invaluable for around-the-clock monitoring.

The YOLOv8 detection framework enhances UAV capabilities by accurately identifying oil patterns and distinguishing them from other elements on the water. Once detected, UAVs can track the spread of the oil slick, providing live data to response teams. This real-time information is vital for deploying resources like containment booms and skimmers to prevent further dispersal of the oil. Additionally, UAVs allow responders to evaluate the extent of an oil spill more effectively than traditional methods, enabling a faster, more coordinated response. In recent case studies, UAVs have been deployed along coastlines and open waters, significantly reducing the time between detection and response, thereby minimizing environmental impact.

Using UAVs for oil spill detection also allows for systematic post-spill assessments. By periodically capturing images of the affected area, UAVs provide a comprehensive view of recovery over time, helping researchers understand the spill's long-term ecological effects. This data is essential for assessing the effectiveness of cleanup efforts and identifying vulnerable regions that require further protective measures.

5.4.2 POLLUTION MONITORING

Pollution monitoring is a critical aspect of environmental conservation, as pollution from plastics, chemicals, and other contaminants poses a significant threat to marine life and habitats. UAVs provide an efficient and scalable solution for monitoring various forms of marine pollution by capturing aerial images and videos over large areas. With object detection capabilities, UAVs can identify and classify pollutants like floating plastic debris, chemical residues, and other visible contaminants, allowing authorities to assess pollution levels accurately.

For instance, UAVs are instrumental in tracking plastic waste that accumulates in specific regions, particularly along coastlines and in "garbage patches" within the ocean. YOLOv8's object detection capabilities allow UAVs to detect individual pieces of plastic or clusters of waste, providing data on pollution density and distribution. This information enables environmental agencies to pinpoint pollution sources and implement targeted clean-up efforts, improving the efficiency of conservation initiatives.

Additionally, UAVs equipped with specialized sensors can monitor chemical pollution by detecting variations in water color or temperature, which may indicate the presence of harmful substances. These sensors, combined with visual detection algorithms, help UAVs identify areas affected by chemical spills or nutrient runoff, which can lead to harmful algal blooms. By monitoring these areas regularly, UAVs provide real-time data that informs pollution management strategies, helping reduce pollution's impact on marine ecosystems.

5.4.3 ECOSYSTEM ASSESSMENT

Ecosystem assessment is essential for understanding the health of marine environments and managing them sustainably. UAVs facilitate this by offering a non-intrusive, detailed perspective on various marine habitats, including coral reefs, seagrass beds, and mangrove forests. Through systematic imaging and data collection, UAVs provide researchers with high-quality data to assess biodiversity, habitat conditions, and the impacts of human activities on these ecosystems.

In coral reefs, for example, UAVs can detect signs of bleaching, disease, and physical damage caused by storms or human interference. The YOLOv8 framework aids in identifying specific features of the reef, such as live coral cover, bleached coral, and algal growth, allowing scientists to quantify the health of the reef over time. UAVs can fly over the same reef area periodically, creating a time-lapse view of ecosystem changes. This longitudinal data enables researchers to assess recovery rates after damaging events and develop strategies to protect the reef from future threats.

In addition to reefs, UAVs provide critical data on other coastal and marine habitats, such as seagrass beds, which serve as nurseries for many marine species, and mangrove forests, which protect shorelines from erosion. By mapping these areas, UAVs help identify changes in vegetation cover, species distribution, and habitat quality. This data is essential for conservation planning, as it allows environmental agencies to prioritize high-risk areas, allocate resources effectively, and measure the success of restoration projects. Regular UAV monitoring of these habitats also enables early detection of environmental stressors, such as pollution or rising water temperatures, which can lead to habitat degradation if left unaddressed.

Overall, UAV-based ecosystem assessment provides a detailed and dynamic view of marine and coastal environments, enabling proactive conservation strategies that adapt to changing environmental conditions. By providing reliable, high-resolution data, UAVs empower conservationists, researchers, and policymakers to make informed decisions that support the sustainable management of marine ecosystems.

Chapter 6

Discussion

6.1 TECHINAL IMPLICATIONS

The deployment of UAV-based tracking systems in marine environments comes with unique technical implications that impact both their performance and their suitability for real-world applications. Our framework, combining YOLOv8 for object detection with DeepSORT for tracking, demonstrates a high level of accuracy and adaptability to dynamic conditions, making it ideal for complex marine scenarios. However, as with any technical solution, there are several factors—algorithm effectiveness, system scalability, and performance trade-offs—that influence the practicality and reliability of UAV systems for environmental monitoring and maritime law enforcement.

6.1.1 ALGORITHM EFFECTIVENESS

Algorithm effectiveness is a critical factor in the success of UAV-based tracking systems, particularly in environments as dynamic and visually complex as marine settings. YOLOv8's role in detecting objects with high precision and recall is essential for identifying small, fast-moving, or partially submerged objects, which are common challenges in ocean surveillance. In tests on datasets like MOBDrone, YOLOv8 achieved impressive metrics in terms of recall and precision, indicating that the algorithm is robust enough to distinguish objects against varying backgrounds, from open water to cluttered coastlines.

The integration of DeepSORT for tracking adds another layer of robustness, enabling the system to assign unique IDs to detected objects and follow their movements over time. This pairing minimizes the likelihood of losing track of targets, even when they are obscured or move quickly across the frame. For example, in applications like anti-poaching and search and rescue, the system's ability to maintain continuous tracking is essential to the success of the mission. The Hungarian algorithm, employed by DeepSORT, optimizes object association by efficiently handling multiple targets, making it highly effective in crowded or busy scenes where numerous objects need to be identified and followed.

However, while YOLOv8 and DeepSORT excel in controlled environments, environmental variables such as water reflections, wave patterns, and changing light conditions can reduce their accuracy. Therefore, further refinements, including adjustments for lighting variations and sensor enhancements like infrared or multispectral imaging, may be necessary to improve detection accuracy in extreme conditions. These refinements would enable the system to maintain high performance across a broader range of scenarios, making it more reliable for comprehensive marine monitoring.

6.1.2 SYSTEM SCALABILITY

Scalability is a major consideration in deploying UAV systems for large-scale operations, as many marine monitoring applications cover extensive and diverse areas. Our system, while highly accurate, faces limitations in terms of processing power, especially when deployed on single UAVs with limited computational resources. To expand the system's scalability, multiple UAVs can be deployed in a coordinated fashion, where each UAV covers a designated segment of the monitored area. This division of labor enables the tracking system to operate over larger areas without compromising its real-time processing capabilities.

Scaling up, however, introduces logistical and technical challenges, particularly in managing the data flow and maintaining consistent performance across multiple UAVs. A coordinated network of UAVs requires a robust communication infrastructure, where each unit can relay data to a central command center or data storage facility. This infrastructure must support high-bandwidth data transmission, as UAVs generate large amounts of image and video data that need to be processed in real time. Integrating edge computing solutions directly onto UAVs could improve scalability by reducing the need for continuous data transmission, as UAVs could process some data locally, only sending critical alerts or processed results back to the command center.

In larger deployments, the addition of cloud computing could further enhance scalability. Cloud-based solutions allow for the centralized processing and analysis of UAV data, enabling operators to monitor broader regions without compromising on detection and tracking quality. This approach also makes it feasible to apply machine learning models to detect new patterns over time, learning from past data to improve accuracy and adapt to changing environmental conditions. Cloud integration thus represents a promising direction for overcoming the scalability challenges associated with real-time marine surveillance, offering a flexible, distributed system for widespread monitoring.

6.1.3 PERFORMANCE TRADE-OFFS

Implementing UAV-based tracking systems inevitably involves performance trade-offs, especially when balancing detection accuracy, processing speed, and power efficiency. In high-stakes scenarios such as search and rescue, where immediate response is essential, the system may prioritize faster detection over maximum accuracy to ensure timely intervention. This trade-off is particularly relevant when operating in areas with rapidly changing conditions, such as turbulent seas or during severe weather events, where a slightly reduced accuracy might be acceptable if it enables faster response times.

One significant trade-off is the balance between computational demand and system efficiency. Algorithms like YOLOv8 and DeepSORT, while effective, require considerable processing power, which can be challenging for UAVs with limited on-board resources. High-resolution image processing, necessary for accurately detecting small objects in vast oceanic settings, places further strain on the system's computing capabilities. To address this, operators may opt for lower-resolution settings in situations where high-speed processing is a priority, accepting a potential decrease in object detection accuracy.

Battery life is another critical factor affected by performance trade-offs. UAVs equipped with high-resolution cameras and multiple sensors tend to have shorter operational times, as energy-intensive processing reduces available flight duration. In applications where continuous surveillance is necessary, this trade-off necessitates a balance between equipment used and operational length. UAVs may be deployed in rotations, with each unit operating in short bursts, allowing for periodic battery recharges while ensuring constant area coverage. Alternatively, operators might deploy UAVs with specialized power-saving settings that reduce image quality slightly to conserve battery life without severely impacting detection capabilities.

Finally, in multi-UAV systems, performance trade-offs also extend to data management, as processing data from multiple sources in real-time requires efficient data fusion techniques. Data fusion enables the system to combine information from multiple UAVs, providing a comprehensive view of the monitored area. However, this process requires advanced software solutions to manage large data volumes without overloading the system, thus balancing the need for real-time insights with the technical limitations of current UAV systems.

6.2 PRACTICAL IMPLICATIONS

Deploying UAV-based tracking and monitoring systems in marine environments brings a range of practical implications that go beyond the technical capabilities of the equipment. To achieve optimal performance, there are important practical factors to consider, including the operational considerations for UAV deployment, the resource requirements for managing UAV fleets, and the various implementation challenges unique to marine environments. Understanding these practical aspects is essential for effective UAV-based monitoring, enabling systems to operate efficiently and reliably in diverse, challenging conditions.

6.2.1 OPERATIONAL CONSIDERATIONS

Operational considerations are central to the effective deployment of UAVs in marine settings, where environmental conditions and mission requirements can vary widely. UAV operations over the open ocean or along coastlines must account for factors such as weather conditions, battery life, flight range, and the complexity of the environment. For instance, UAVs deployed for search and rescue missions often need to navigate high winds, sudden weather changes, and ocean spray, all of which can affect flight stability, visibility, and sensor performance.

Another key operational factor is the flight range and duration of UAVs. UAVs used in marine applications are typically battery-powered, which limits their operational time, especially when equipped with high-resolution cameras and other power-intensive sensors. Extended flight times are crucial for missions that require broad area coverage or involve long-term monitoring, such as tracking the movement of vessels or observing marine wildlife. One solution is to deploy UAVs in rotations, where one UAV begins its mission as another returns to base for recharging, ensuring continuous coverage. Alternatively, solar-powered UAVs, though less common, could extend operational times, especially in sunny coastal regions, by supplementing battery power with solar energy.

Operational considerations also extend to navigation and communication. UAVs require reliable GPS and communication links to maintain accurate positioning and transmit data back to command centers. In remote or offshore locations, establishing consistent communication channels can be challenging, particularly when UAVs are operating beyond the line of sight. Solutions like satellite communication or the use of dedicated relay UAVs can enhance data transfer capabilities over long distances, though these add complexity and cost to operations. Without such infrastructure, UAVs may struggle to relay real-time data, impacting mission effectiveness.

6.2.2 RESOURCE REQUIREMENTS

The successful deployment and operation of require significant resource investment. This includes not only the physical hardware, such as UAVs equipped with high-quality sensors and tracking systems, but also the software infrastructure for processing and analyzing the collected data. High-resolution cameras, thermal sensors, and advanced onboard computing resources are essential for detecting and tracking objects accurately in marine environments, where visibility and lighting conditions can vary significantly. Furthermore, resource requirements for UAV fleets increase with the complexity of the mission and the scale of the area to be monitored.

In addition to hardware, skilled personnel are necessary for UAV deployment, maintenance, and data interpretation. UAV operators require specialized training to handle marine-specific challenges, such as maintaining flight stability over water and adjusting for wind and salt exposure. Technicians are also essential for conducting routine maintenance on UAVs, especially in marine settings where saltwater corrosion can damage equipment. Data analysts play a critical role in interpreting the information collected by UAVs, transforming raw data into actionable insights for search and rescue teams, law enforcement, or conservation agencies.

Moreover, cloud storage and data processing resources are required for managing the large volumes of video and image data generated by UAVs. In extensive monitoring operations, especially those involving multi-UAV networks, these systems can quickly generate terabytes of data, which need to be stored, processed, and analyzed in real time. Cloud-based solutions are often used to provide scalable data storage and processing power, enabling real-time analysis and long-term data archiving. However, cloud infrastructure adds to operational costs and requires robust cybersecurity measures to protect sensitive information. Data transfer costs also rise when transmitting high-resolution imagery, making it essential to budget for both storage and communication resources when deploying UAV fleets for marine applications.

6.2.3 IMPLEMENTATION CHALLENGES

The implementation of UAV systems in marine environments comes with unique challenges due to the harsh and unpredictable nature of these settings. One of the primary challenges is weather dependency, as UAV operations are highly susceptible to changes in weather. Strong winds, heavy rain, and salt spray can disrupt flights, damage sensors, and reduce visibility, limiting UAV functionality. Additionally, saltwater corrosion is a significant concern, as prolonged exposure can degrade UAV components over time. Regular maintenance, protective coatings, and corrosion-resistant materials are essential to extend the lifespan of UAVs operating in marine environments, although these measures increase overall operational costs.

Another implementation challenge is regulatory compliance. UAV operations, particularly in coastal and international waters, are subject to various regulations, including airspace restrictions, privacy laws, and maritime safety standards. Navigating these regulatory frameworks can be complex, especially in areas where national jurisdictions overlap or where UAV activities may disturb local communities or wildlife. Compliance with these regulations often requires obtaining special permits or licenses, which can be time-consuming and require careful coordination with authorities. Additionally, regulatory restrictions on UAV flight height, operational range, and payload capacities can limit the flexibility and effectiveness of UAV missions.

Data management and integration present additional challenges in implementation. UAVs generate vast amounts of data, which must be transmitted, processed, and stored efficiently to provide real-time insights. In remote locations, ensuring uninterrupted data transfer and storage can be difficult, especially when cloud-based solutions are limited by connectivity issues. Moreover, integrating data from multiple UAVs in large-scale operations requires advanced data fusion techniques to create a cohesive view of the monitored area. This integration is crucial for applications like vessel monitoring or environmental assessments but necessitates robust software solutions capable of handling high data volumes and resolving inconsistencies between multiple data sources.

Finally, cybersecurity is a growing concern in UAV operations, particularly for systems that rely on cloud storage or transmit sensitive data. Unauthorized access to UAV data can pose security risks, especially in law enforcement or environmental monitoring applications where data confidentiality is critical. Cybersecurity measures, including encryption, secure communication channels, and access controls, are necessary to protect UAV data from interception or tampering. Implementing these safeguards requires additional investment and technical expertise, making it a significant consideration in the planning and execution of UAV-based marine monitoring projects.

6.3 LIMITATIONS AND CHALLENGES

While UAVs have proven effective for marine monitoring and tracking applications, their deployment also comes with significant limitations and challenges that impact their performance and reliability. These challenges encompass technical constraints related to hardware and software, environmental factors that affect operational stability, and various operational issues that complicate mission execution. Addressing these limitations is critical to enhancing the utility and adaptability of UAVs in marine environments.

6.3.1 TECHNICAL CONSTRAINTS

Technical constraints represent a primary challenge in UAV operations for marine applications, especially due to the limitations of hardware and software capabilities. UAVs rely heavily on battery power, which restricts their flight duration and operational range. This limitation is particularly problematic for extended missions, such as continuous monitoring or search and rescue, where longer flight times are required. While some UAVs can be equipped with larger batteries or alternative power sources like solar panels, these solutions add weight and cost.

Another technical constraint is the processing power available on UAVs, which directly affects their ability to perform complex computations, such as real-time object detection and tracking. YOLOv8 and DeepSORT, for example, require significant processing resources to achieve high accuracy in detecting and tracking objects, especially in high-resolution imagery. For UAVs with limited on-board processing power, this can lead to slower detection times, reduced accuracy, or an increased need to transmit data to a remote server for processing, which may introduce latency. Additionally, the storage capacity of UAVs is often limited, which restricts the amount of data they can collect on a single flight, especially in missions requiring high-resolution video or image capture.

Sensor limitations also present technical challenges. UAVs deployed for marine monitoring rely on visual, thermal, or multispectral sensors, each with its advantages and limitations. Visual sensors are effective in clear conditions but struggle in low light or fog. Thermal sensors, while helpful in identifying heat signatures, are less useful during daytime over the ocean, where the sun's heat affects accuracy. Multispectral sensors provide additional environmental insights but are costly and require more complex data processing, which may not be feasible on standard UAV platforms. These sensor limitations highlight the need for multi-sensor integration and advanced processing solutions to optimize detection accuracy across diverse conditions.

6.3.2 ENVIRONMENTAL FACTORS

Environmental factors are a major challenge for UAV operations in marine settings, as they directly impact flight stability, sensor accuracy, and overall mission success. Marine environments are characterized by unpredictable weather patterns, including high winds, storms, and saltwater exposure, all of which can interfere with UAV functionality. Strong winds, for example, can destabilize UAVs, making it difficult to capture steady images or maintain precise location tracking. In extreme cases, turbulent conditions may even result in UAV crashes, risking the loss of expensive equipment and potentially endangering marine life or habitats.

Saltwater exposure poses another significant challenge. Prolonged exposure to saltwater and moisture can corrode UAV components, affecting performance and reducing the system's operational lifespan. UAVs operating in marine environments require special protective coatings or materials resistant to corrosion, which adds to maintenance and operational costs. Additionally, regular maintenance is essential to prevent salt damage, as even brief exposure to saltwater can deteriorate key components like propellers, joints, and electrical systems.

Visibility issues also complicate UAV operations in marine settings. Glare from the sun, reflective water surfaces, fog, and low-light conditions can reduce the effectiveness of visual sensors, making it difficult for detection algorithms to accurately identify and track objects. For example, reflections on the water can create false positives in object detection, confusing the system and reducing detection accuracy. To mitigate this, UAVs may require advanced filters or the integration of complementary sensors, such as LiDAR or radar, though these additions increase system complexity and cost. UAV operators must also carefully time their missions to avoid unfavorable lighting conditions, which can limit operational flexibility.

Temperature fluctuations present another environmental challenge. UAVs are designed to operate within certain temperature ranges, but exposure to extreme heat or cold can affect battery life, processing speeds, and sensor functionality. Marine environments, especially tropical or polar regions, experience temperature extremes that can strain UAV systems. Extreme heat, for example, can reduce battery efficiency and lead to overheating, while cold weather can reduce battery performance and affect sensor calibration. Addressing these environmental factors requires specialized equipment and careful mission planning, though complete mitigation is often not feasible.

6.3.3 OPERATIONAL ISSUES

Operational issues further complicate the deployment of UAV systems in marine applications, as they involve logistical, regulatory, and coordination challenges. One major operational limitation is the restricted flight time due to battery life. Given that marine missions often require wide-area surveillance or extended tracking, the limited battery life of UAVs necessitates frequent battery replacements or rotations of multiple UAVs to maintain continuous coverage. This constant need for recharging or replacing batteries adds complexity to mission planning and increases operational costs.

Data management is another operational issue. UAVs generate substantial amounts of data, particularly in missions that involve high-resolution imaging or video recording. Transmitting and storing this data can be challenging, especially in remote marine locations with limited connectivity. For real-time applications, such as search and rescue, delays in data transmission can reduce the system's effectiveness. Satellite communication or data relay UAVs may be used to overcome connectivity issues, but these options add to operational complexity and cost. Additionally, data storage becomes a concern in long-term monitoring missions, as UAVs require sufficient storage capacity to collect data continuously, which may require additional investments in cloud storage or data management infrastructure.

Regulatory compliance also poses operational challenges. UAVs operating over marine environments must comply with local and international regulations, particularly in areas where multiple jurisdictions overlap. For example, UAVs used for law enforcement or surveillance near national borders may require special permissions, and operators may need to coordinate with different regulatory agencies. Regulations concerning UAV flight height, operational range, and data privacy vary by country and can restrict the flexibility of UAV missions. Ensuring compliance with these regulations requires significant administrative effort and can limit the range and scope of UAV operations, particularly in sensitive areas like marine protected zones.

Finally, operator training and expertise are essential for the effective deployment of UAV systems but can be resource-intensive. Operators must be skilled in handling UAVs in marine environments, where sudden weather changes, salt exposure, and navigation challenges demand quick decision-making and technical proficiency. Furthermore, operators must be trained in data interpretation and real-time decision-making, as UAV missions often involve time-sensitive tasks like identifying threats or locating people in distress. Maintaining a trained team of UAV operators and support staff requires ongoing investment in training and resources, impacting the overall cost and feasibility of large-scale UAV operations.

6.4 FUTURE IMPROVEMENTS

While UAV-based tracking and monitoring systems have shown great promise in marine applications, further improvements are essential to fully unlock their potential. Enhancements in algorithms, system optimizations, and additional features can help address current limitations, improve efficiency, and expand the range of applications. These future improvements aim to make UAV systems more robust, accurate, and adaptable to the challenges of marine environments.

6.4.1 ALGORITHM ENHANCEMENTS

Algorithm enhancements are vital for increasing the accuracy and reliability of UAV-based tracking systems in dynamic and challenging marine settings. Currently, detection algorithms like YOLOv8 perform well in distinguishing objects from complex backgrounds, but further development could improve their robustness in varied environmental conditions, such as low light, fog, and high reflection from the water surface. Future algorithms could incorporate adaptive learning techniques that adjust parameters in real-time based on environmental factors, ensuring higher accuracy under fluctuating conditions.

In addition, incorporating artificial intelligence (AI) models that utilize deep learning frameworks could enable UAVs to identify a broader range of objects with greater precision. For instance, algorithms that combine image recognition with contextual awareness could allow UAVs to differentiate between similar objects, such as identifying specific species of marine animals or types of vessels based on subtle visual cues. This capability would be especially useful for applications like species monitoring or vessel identification in law enforcement. Algorithm advancements could also include improved motion prediction, enabling more accurate tracking of fast-moving or evasive objects, such as poachers or fleeing vessels.

Integrating multi-sensor data fusion into the algorithm design is another area for enhancement. By combining data from visual, thermal, and multispectral sensors, future algorithms can provide a more comprehensive view of the environment, leading to higher detection accuracy. Multi-sensor fusion can also mitigate challenges associated with single-sensor limitations, such as poor visibility or thermal interference, creating a robust framework that adapts seamlessly to varying conditions. With enhanced algorithms, UAV-based systems could deliver superior performance in applications ranging from search and rescue to environmental monitoring.

6.4.2 SYSTEM OPTIMIZATIONS

System optimizations are crucial for enhancing the performance, efficiency, and scalability of UAV systems, especially in resource-constrained environments. One key area for optimization is energy management. Limited battery life remains a significant challenge, restricting flight time and operational range. Future UAVs could benefit from advanced power management systems that dynamically adjust energy use based on mission priorities. For example, when operating in remote areas, UAVs could switch to energy-saving modes during idle periods or low-risk monitoring phases, conserving power for high-priority tasks like search and rescue.

Optimizations in data processing could further improve system performance. UAVs typically process large volumes of visual data, and implementing edge computing solutions directly on the UAV could reduce the need for continuous data transmission to remote servers. By processing data locally, UAVs can perform real-time analysis, reducing latency and enhancing decision-making in time-sensitive missions. As edge computing technology advances, UAVs could handle increasingly complex computations on-board, allowing for faster detection, tracking, and alerting capabilities, even in remote areas with limited connectivity.

Communication optimization is also essential for multi-UAV operations. In larger deployments, where multiple UAVs monitor expansive areas simultaneously, efficient communication networks are critical. Implementing mesh network architectures could enable UAVs to communicate directly with each other, sharing data and coordinating tasks in real-time without relying solely on ground stations. This approach improves coverage and resilience, as each UAV serves as a relay point, extending the network range and ensuring consistent data flow. Mesh networks also allow UAV fleets to adapt dynamically to mission demands, reallocating resources and adjusting positions as needed to maintain optimal coverage.

6.4.3 FEATURE ADDITIONS

Incorporating new features into UAV systems would significantly enhance their capabilities for marine monitoring, enforcement, and conservation. One promising addition is advanced sensors, including LiDAR, hyperspectral, and sonar sensors. LiDAR, for instance, could improve object detection accuracy in low-visibility conditions and provide detailed 3D maps of the environment, valuable for both search and rescue and environmental assessments. Hyperspectral sensors could be used to detect chemical compositions in the water, helping identify pollutants or oil spills, while sonar sensors would enable underwater mapping and tracking, opening up new possibilities for monitoring underwater activity and marine life.

Another valuable feature is autonomous decision-making, allowing UAVs to perform tasks with minimal human intervention. By integrating machine learning models that analyze patterns in real time, UAVs could make autonomous adjustments based on mission needs. For instance, a UAV conducting species monitoring could adjust its flight path when detecting increased animal activity, focusing on areas with the highest data relevance. In law enforcement applications, autonomous UAVs could follow suspicious vessels, capturing continuous footage without requiring manual control. Autonomous decision-making would not only improve operational efficiency but also reduce the workload on human operators.

The addition of night vision capabilities would further expand the operational scope of UAVs, allowing them to operate effectively at all times. Currently, UAVs rely on daylight or thermal imaging, which has limitations in low-light conditions. Advanced night vision cameras, capable of capturing high-resolution imagery in dark conditions, would enhance UAV functionality in applications that require around-the-clock monitoring, such as anti-poaching patrols and emergency response. Coupling night vision with AI-based detection algorithms could allow UAVs to perform detailed surveillance even in complete darkness, greatly enhancing their utility in marine and coastal security.

Lastly, implementing real-time data analytics would add substantial value to UAV-based monitoring systems. By incorporating real-time analytics, UAVs could detect and flag unusual patterns or behaviors as they occur, providing immediate alerts to authorities or conservation teams. For example, in environmental monitoring, UAVs could identify and notify operators about abrupt changes in water color that may indicate an oil spill, or detect sudden movements in marine wildlife that could suggest distress. Real-time analytics not only improve response times but also provide richer data insights, helping teams make data-driven decisions on the spot.

Chapter 7

Conclusion and Future Work

This project aimed to advance the accuracy and efficiency of marine surveillance by integrating object detection and tracking models, specifically YOLO for detection and DeepSORT for tracking. A key success of the project lies in its development of a UAV-based surveillance model capable of recognizing and tracking marine objects with promising initial accuracy. The primary accomplishments include establishing a framework that could effectively detect and track objects in aerial footage, enhancing real-time monitoring potential for marine conservation and rescue applications. Yet, this journey was not without challenges, such as difficulties in training the YOLO model to handle complex marine environments, integration hurdles with DeepSORT, and addressing accuracy inconsistencies that affected tracking precision. These challenges led to critical adjustments, providing insights into the fine-tuning requirements for real-world application.

The approach of combining YOLO with DeepSORT offered valuable insights into the strengths and limitations of deep learning in dynamic settings. Notably, the project highlighted how factors like environmental variations, water reflection, and object occlusion impacted detection accuracy. These issues underlined the importance of adapting machine learning frameworks to the specific conditions they aim to monitor, a consideration that will be essential in future developments. From an academic perspective, the project met key objectives, including the successful deployment of a functional tracking model, yet identified technical constraints that could be improved to achieve the intended real-time accuracy and responsiveness.

Given additional time and resources, further research could focus on refining model accuracy through hyperparameter tuning and leveraging more diverse training datasets that represent a broader range of marine conditions. Additional experimentation with alternative tracking algorithms or a hybrid tracking model could enhance tracking reliability in challenging scenarios. Performance improvements might also be achieved by integrating adaptive algorithms that adjust for environmental shifts, such as lighting changes and wave patterns, ensuring a more robust surveillance capability.

The main takeaway from this project is the feasibility of UAV-based marine surveillance systems, paired with a clear understanding of the technical adaptations required to achieve operational reliability. While the initial framework demonstrates substantial benefits, including the potential for real-time conservation and rescue applications, much remains to be done in optimizing accuracy and scalability. The project lays a foundation for future work, with a well-defined path toward creating a refined, high-precision marine monitoring solution capable of tackling real-world complexities.

METHODOLOGY	RECALL	PRECISION	F-SCORE
YOLOX+ByteTracker	83.5	89.3	86.8
YOLOX-XL+ OCSORT	85.2	90.1	87.6
YOLOv7+MoveSORT	86.7	90.9	88.8
YOLOv7+StrongerSORT	87.1	91.2	89.1
YOLOv8+DeepS-ORT	88.7	92.9	90.75

Table1. Comparative analysis of various models

In conclusion, the use of UAVs for maritime surveillance and rescue operations, including species monitoring, anti-poaching, and human life rescue, requires highly accurate object detection and tracking. The proposed framework combines YOLOv8 for object detection and DeepSORT for object tracking and association, utilizing the Hungarian algorithm and Kalman filter to enhance tracking accuracy. The C2f module, which integrates high-quality features and contextual data, further improves detection performance. Tested on the MOBDrone dataset, this framework outperforms other deep learning-based models, achieving a Recall of 88.7%, a Precision of 92.9%, and an F-Score of 90.75%. These results demonstrate the framework's effectiveness in addressing the challenges of marine life applications and other related domains.

Appendices

Appendix 1: Dataset and Preprocessing

This appendix covers the datasets used, including MOBDrone along with a brief overview of preprocessing techniques. Sample images and data augmentation methods are provided to illustrate how data was prepared to improve model robustness.

Appendix 2: Model Configuration and Experimental Setup

Outlines key configurations for YOLOv5 and DeepSORT, including primary hyperparameters (e.g., learning rate, batch size) and tracking metrics. Also provides details on the experimental setup, specifying the hardware (NVIDIA RTX 4050 GPU) and software environments (Python, PyTorch) to ensure reproducibility.

Appendix 3: Results and Error Analysis

Presents additional performance metrics, such as precision, recall, and F1 scores under different visibility conditions, along with screenshots showcasing successful and challenging cases. Error analysis addresses common issues, including object occlusion and misclassification, with notes on potential improvements.

Appendix 5: Code

```
import re

from pathlib import Path

import pkg_resources as pkg

from setuptools import find_packages, setup

# Settings

FILE = Path(__file__).resolve()

ROOT = FILE.parent  # root directory

README = (ROOT / "README.md").read_text(encoding="utf-8")

REQUIREMENTS = [f'{x.name}{x.specifier}' for x in
pkg.parse_requirements((ROOT / 'requirements.txt').read_text())]

def get_version():

    file = ROOT / 'ultralytics/__init__.py'

    return re.search(r'^__version__ = [\'"]([^\'"]*)[\'"]', file.read_text(), re.M)[1]

setup(

    name="ultralytics",  # name of pypi package
```

```
    version=get_version(),  # version of pypi package
    python_requires=">=3.7.0",
    license='GPL-3.0',
    description='Ultralytics YOLOv8 and HUB',
    long_description=README,
    long_description_content_type="text/markdown",
    url="https://github.com/ultralytics/ultralytics",
    project_urls={
        'Bug Reports': 'https://github.com/ultralytics/ultralytics/issues',
        'Funding': 'https://ultralytics.com',
        'Source': 'https://github.com/ultralytics/ultralytics',},
    author="Ultralytics"
    author_email='hello@ultralytics.com',
    packages=find_packages(),  # required
    include_package_data=True,
    install_requires=REQUIREMENTS,
    extras_require={
        'dev':
        ['check-manifest', 'pytest', 'pytest-cov', 'coverage', 'mkdocs',
'mkdocstrings[python]', 'mkdocs-material'],},
    classifiers=[
        "Intended Audience :: Developers", "Intended Audience :: Science/Research",
        "License :: OSI Approved :: GNU General Public License v3 (GPLv3)",
"Programming Language :: Python :: 3",
        "Programming Language :: Python :: 3.7", "Programming Language :: Python ::
3.8",
        "Programming Language :: Python :: 3.9", "Programming Language :: Python ::
3.10",
```

```
"Topic :: Software Development", "Topic :: Scientific/Engineering",
"Topic :: Scientific/Engineering :: Artificial Intelligence",
"Topic :: Scientific/Engineering :: Image Recognition", "Operating System :: POSIX :: Linux",
"Operating System :: MacOS", "Operating System :: Microsoft :: Windows"],
keywords="machine-learning, deep-learning, vision, ML, DL, AI, YOLO, YOLOv3, YOLOv5, YOLOv8, HUB, Ultralytics",
entry_points={
'console_scripts': ['yolo = ultralytics.yolo.cli:cli', 'ultralytics = ultralytics.yolo.cli:cli'],})
```

Main yolo file:

```
from IPython.display import HTML
from base64 import b64encode
import os
# Input video path
save_path = '/content/YOLOv8-DeepSORT-Object-Tracking/runs/detect/train/test1.mp4
# Compressed video path
compressed_path = "/content/result_compressed.mp4"
os.system(f"ffmpeg -i {save_path} -vcodec libx264 {compressed_path}")
# Show video
mp4 = open(compressed_path,'rb').read()
data_url = "data:video/mp4;base64," + b64encode(mp4).decode()
HTML("""
<video width=400 controls>
    <source src="%s" type="video/mp4">
</video>
```

```
from IPython.display import HTML

from base64 import b64encode

import os


# Input video path

save_path = '/content/YOLOv8-DeepSORT-Object-Tracking/runs/detect/train2/test3.mp4'

# Compressed video path

compressed_path = "/content/result_compressed.mp4"

os.system(f"ffmpeg -i {save_path} -vcodec libx264 {compressed_path}")

# Show video

mp4 = open(compressed_path,'rb').read()

data_url = "data:video/mp4;base64," + b64encode(mp4).decode()

HTML("""

<video width=400 controls>

    <source src="%s" type="video/mp4">

</video>

""" % data_url)
```

Appendix 4: Output

Figure 4: Object tracking and association in MOBDrone dataset

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